

E-GOVERNMENT INITIATIVES IN NEPAL

Almost two decades after the first initiative towards e-Governance through the IT policy of 2000, Nepal has so far struggled with implementation of e-Governance due to political instability and the digital divide caused by socio-economic issues such as lack of infrastructure, low-income level and low-literacy rate among others. However, this case study showed that with trend towards foreign labour, Nepal has seen a huge demand for communication resulting in improvement in the communication infrastructure. With access to mobile technology, possibility of citizen reach and interaction through the means of mobile communication has become realizable.

Cyber Laws

Cyber laws are fundamental. They provide security to not only the intellectual property of IT companies but also helps to maintain the privacy of internet users. They check the programs of corporates to make the internet a neutral platform, help to create standard models of use which helps to create tailored facilities to the citizens to boost the economy, and so on.

Different countries have different cyber laws and cyber laws regulating bodies. In Nepal cyber law is called as Electronic Transaction Act (ETA) 2063, which was passed in 2004. The bill deals with issues related to digital signature, intellectual property, cybercrime, etc. The Act is divided into 12 sections and 80 clauses. This law keeps an eye on issues which are related to computer networks and cybercrime. It brings cyber criminals under the justice of law and penalizes them just like other crimes. As per the Act, if anyone is found violating cybercrime, he/she will be punished for a minimum of 6 months to a maximum of 3 years in jail and has to pay minimum 50 thousand to maximum three lakhs as a penalty.

The Electronic Transactions Act, 2063 (2008)

1. This Act may be called “The Electronic Transactions act, 2063 (2008)”.
2. This Act shall be deemed to have been commenced from 24 Bhadra 2063 (sep.2, 2006).
3. This Act shall extend throughout Nepal and shall also apply to any person residing anywhere by committing an offence in contravention to this Act.

What is electronic Transaction?

The electronic transaction is :

- transactions of electronic records data by using any types of electronic means.
- contains electric records and valid digital medium.
- the exchange of all types of records which are in the form of electronic.

Objectives of the Electronic Transaction Act 2063

1. To make legal provision for authentication and regulation of electronic data.
2. To make a reliable date generation, communication, and transmission.
3. To make a secured and authentic means of electronic communication.

4. To regulate all the relating matters of electronic transactions.

Scopes of the Electronic Transaction Act 2063

1. Creation and use of digital signature
2. Control cyber/computer-related crimes.
3. Protection of intellectual property.
4. Protection of confidentiality.
5. Regulations of an electronic transaction by establishing regulating bodies – such as
 - Office of Certificate Control (OCC)
 - Certificate Agencies (CA)
 - Subscribers

Major provisions of the Electronic Transaction Act 2063

1. It has the provision relating to electronic records and digital signature.
2. It has the provision relating to dispatch, receive an acknowledgment of electronic records.
3. It has the provision of few regulating bodies and their functions, rights, and duties such as –

Office of Certificate Control (OCC)

- Certification Agencies (CA)
 - Subscriber
4. It has the provision relating the use of digital signature and certificate.
 5. It has the provision relating to government use of the digital signature.
 6. It has a provision relating to the computer network and network services providers.
 7. It has the provision relating to computer-related crimes and punishments.
 8. Provision of IT tribunal is defined as the first jurisdictional and appellate jurisdiction. IT tribunal (□□□□□ □□□□□□□ □□□□□□□□□□□)
 - First Jurisdictional (□□□□□ □□□□□□□□□□□□□)
 - Appellate Jurisdiction (□□□□□□□□□□ □□□□□□□□□□□□□)
 9. It defines a few documents which this act is not applicable for
 - All the negotiable instrument defined by the “Negotiable Instrument Act 2034”
 - All the documents related to the ownership of properties (dhanipurja, laalpurja ect).
 - Any documents which are used as a primary document in the court process

Cyber crime

Cyber crime is a crime committed by using cyber means/computer technology. Cyber crime is also known as computer-related crime. All the illegal activities committed by using or with the application as computer technology such as –

- Damage to computer and computer system.
- Acts to gain illegal access into the system.
- Use as weapons to commit other crimes.
- Acts against the provision of cyber law.

Some examples of cyber/computer crimes are

- Creating fake documents
- Blackmailing/social media abuse
- Distribution of pornography materials
- Misuse of electronic cards
- Piracy of intellectual properties
- Spreading computer viruses/malware

Major computer-related crimes and punishments

Pirate/ Destroy/Alter computer source code

Unauthorized access in the computer system

Damage to computer and computer system

- Up to 3 years of imprisonment, or up to 2 lakh fine or both

Distribution of electronic materials in electronic form – Not to publish the prohibited materials which are prohibited by prevailing laws, the materials which may create jeopardy in social harmony, spread hate and jealousy, materials which may create harassment to women trafficking, pornography materials

- Up to 5 years imprisonment or up to 1 lakh or both

Disclose of Confidentiality

Deform false statement by parties at the time of the electronic transaction

- Up to 2 years imprisonment or 1 lakh fine or both

Display and submit false license for electronic means

- Only display 1 lakh fine
- Display+submit and perform = 2 years imprisonment or 1 lakh fine or both

Non-submission of prescribed statements

- Up to 50,000 fine

Computer fraud

- Compensate claim amount
- Fine up to 1 lakh and 2 years of imprisonment or both

Computer fraud

- For abetment fine up to 50,000 and up to 6 months imprisonment

For accomplice

- Half of the main convicted

Implementation in the Land Reform

Land reforms played a key role in the national economic development of those countries. Land reforms paved the way for agricultural development, which in turn led to industrial development in these countries. Nepal has passed through a similar phase, too.

Since the late 1950s governing administrations of all political hues have pronounced land to the tiller redistribution necessary to abolish feudalism and advance economic transformation. And yet they have as recurrently failed to achieve these objectives. Although overlord powers of landlords and colonizing agents have been successfully diminished, other destructive characteristics of feudal land relations continue to flourish. These include high levels of absentee landlordism, large areas of underutilized farmland, and failure to reform the conditions of those who farm others' land, some of whom, extraordinarily, remain in debt peonage until today and despite repeated legislation against this. Insecurity of tenure afflicts the entire farming sector, a main cause of its stagnation. Farming has become one of the least appealing ways to survive and yet continues (and will of necessity continue) to underwrite the economy. This is itself truncated through failures to restructure farm tenure relations and force idle landlord capital into off-farm productive enterprise. Breaking out of this cycle has become paramount to post-conflict and now republican New Nepal. Recommitment to abolition of feudalism through land reform has been given declamatory constitutional force. Reform is now termed scientific land reform, seemingly to reflect what is fast emerging as its dominant focus upon the commercialization of agriculture.

Land Reform in Nepal – Where is it coming from and where is it going?

Overview

This is a summary of a book which reports upon a review of land reform in Nepal. The book finds that state-led reformism is far from new in Nepal. Since the late 1950s governing administrations of all political hues have pronounced land to the tiller redistribution necessary to abolish feudalism and advance economic transformation. And yet they have as recurrently failed to achieve these objectives. Although overlord powers of landlords and colonizing agents have been successfully diminished, other destructive characteristics of feudal land relations continue to flourish. These include high levels of absentee landlordism, large areas of underutilized farmland, and failure to reform the conditions of those who farm others' land, some of whom, extraordinarily, remain in debt peonage until

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Limitations on farm size or absentee landlordism are no longer mentioned and taxation (inducing sales to better-off farmers) is offered as the mechanism to reduce idle lands. The heat is to be taken out of popular demand for land by providing tiny plots of public wasteland to groups of the ultra-poor, and to those who have voted with their feet time and time again over the last 50 years and once again begun to occupy public lands in organized squatter camps. This strategy has ominous echoes of former decades. It enables the State to avoid tackling the founding issues of unjustly acquired properties in the private sector and the rampant exploitation which continues to affect up to 2.8 million households who either have no shelter or farm of their own or not enough land to even subsist from – and while 7.5% of rural households still retain over a third of the cultivated land area, and a much greater proportion of the private property estate overall, significant parts of it idle. It also enables the State to sidestep the increasingly pressing question as to how it has come to be the largest landholder in Nepal in the course of reforms, a situation at least partly engineered on the back of classical abolition of customary land rights in the name of modernization, and the co-option of naturally collectively owned assets like forests and pasturelands to its own purse and presumed superior guardianship. This has been a strategy upon which it has wisely but as yet highly partially back-tracked, in the form of handing over managerial control of mainly degraded forests to community bodies. In the process it has set up a new set of awkward institutional precedents which inhibit full restitution of property rights to either customary or modern community possession, and which trigger increasingly tense contestation over rights at the local level and especially between indigenous and non-indigenous communities. This is however unlikely to be the end of the story. Although often un-crystallized in its vision, public demand for reform seems to heighten in this post-conflict period. This looks to real reform not just in the pivotal landlord-tenant relationship but in State-people relations, in terms of both power over property and the status of public lands. Meanwhile a High Level Commission is to be formed to produce concrete recommendations for land reform. A new Constitution is to be drafted in 2009 providing another opportunity for a fresh approach to property matters. How far they will be able or enabled to take up the challenge of reforming reform remains to be seen. The risks of not doing so are high, with weary return to interventions of old likely, but with promises that this will work better this time on the basis of anticipated stronger political will. Indeed, lack of political will is identified by this study as historically the outstanding impediment to successful redistributive reform around the world. Nonetheless, too much reliance upon this is unreliable in a modern democratic state and within one which is so vibrantly Party-riven and where the age-old alliance of (often absentee) landlordism and officialdom appears to remain surprisingly intact. That is, sufficiently robust political will may be difficult to achieve now as in the past.

Summary

Democratizing land reform itself offers a better chance of success, in the sense of looking to public will. This requires structuring reform as a people's program in which every village community is empowered to control its own land relations, within the parameters of broad national principles and more exacting inclusive process. This offers a more workable route out of the quagmire of broken promises, breakdown in the rule of land-related law, and a narrowness of objective and remedy than back to business as usual with some tinkering can achieve. Correctly empowered communities will have a better chance of breaking the chains of the embedded landlord-State allegiances which limit change. With control over their own community land registers, they will be better able to ensure the long-term occupancy and rights of each household are registered and disputes around these more quickly resolved. A community based approach, working in an incremental manner, and on an area by area basis, will also be better able to tackle the rights and governance grievances which embrace much of the public land estate, including existing and future protected areas. Relieved of powers to manage land distribution itself and the opportunities for rent-seeking which afflict current systems, district authorities will be better able to neutrally facilitate delivery and monitor, with accountability systems firmly embedded in local populations, not upwards to central government. Taking into account the experiences of innovative community based approach to land rights reform elsewhere, we do not find it too fanciful to envision a thriving future Nepal as a mosaic of some 4,000 or more collectively-governed community land areas embracing the entire rural landscape; and within which elected Community Land Boards govern not only private land holding but the regulation, use and management of collectively owned resources including pasturelands and protected forests and reserves, under the oversight of national or federal legislation and technical agencies. How far emergent reformism can extend its vision and scope in such ways (and similarly embrace the equally pressing issues around rapid urban growth and speculative land hoarding and housing developments once again to the detriment of the poor, in especially peri-urban villages, not covered in this volume) remains to be seen. Ironically, intensive politicization along party lines may yet serve to limit the level of community-driven empowerment needed to drive real change in this areas. In the interim, every assistance should be afforded the debate and policy-makers to maximise the current opportunity to genuinely move forward.

Nepal has performed poorly

These outputs compare poorly with redistributive reform around the world. The better of these have seen more than 60% of rural households benefit (e.g. Vietnam, China, Taiwan, Japan, Korea, Cuba, Ethiopia, Mexico, Russia, Armenia) and deliver land to nearly a billion beneficiaries overall. However, the results are not worse than for the Indian sub-continent as a whole, excepting the Indian state of West Bengal. Failure to reform the farm-based economy has failed transformation overall. Meanwhile the uncertainties created by poorly enforced reforms have played a main role in inhibiting agricultural development and prosperity. Half a century after the first significant land reform legislation (1957) neither land owners nor tenants and workers have clear and stable control over the founding means of production – the land. Out-migration from agriculture has become the major escape route but largely excluding the most poor. Agriculture itself stagnates. Perhaps worse, failure to perform, or misconstrued reform, have often made things worse for the disadvantaged majority. Even 'successes' have their pyrrhic underside. For example, a main output of imposing ceilings was concealment

of ownership and severe erosion of the integrity of records. A main effect of tenancy registration was eviction; a main effect of taking uncultivated land from the rich was dispossession of the poor, and a main effect of presumed modernization of tenure forms was characteristic abuse of customary property rights. Failure to perform overall contributed to civil war, land reform becoming a clarion call of the Maoist insurgency.

The institutional basis of feudalism has been weakened but its subjects not yet liberated. As public documents acknowledge, elements of feudalism continue to exist in 2008. These manifest most painfully in the persistence of serfdom, despite repeated legal outlaw of such conditions (in fact first in 1924). More widely there is unjustified rife indebtedness in the rural sector, caused by sustained exploitation of tenancy and worker arrangements, and gross insufficiency of land of their own to farm. Nor has absentee landlordism been removed. This is integral to a continuing economic malaise through which better-off owners still gain more benefit from underpaying labour to produce food crops than from investing in the farm (and labour) to increase, improve, and diversify production.

Land reform did succeed in undercutting the powerbase of feudal property relations as taxers and controllers of land access. However the powers these characteristic feudal zamindari (talukdar and jimidar) lost became powers of the central state, not ordinary citizens.

Nationalization of collective assets has been a questionable output of the reforms. Reforms have also seen the state emerge as the majority landholder through nationalization of forests, wastelands and pastures and the abolition of customary property rights affecting those resources. This too was to the loss of the majority poor. While this has proven a common thread in so-called modernization of land relations globally including post-feudal and post-colonial reformism, reversals are ultimately proving necessary.

Modernization has in respects aided polarization. The earliest and arguably most successful thrust of land reform from the 1950s was to consolidate holdings as fungible private property, whereas prior to this, many lands were held to be easily revertible to King and State without compensation. Without protection of tenants or labour included in this process, this too played its role in diminishing the rights and security of the majority, by finally sealing the gap between overlords and tillers; the former became registered land owners, the latter, mere tenants.

Poorly seen-through tenancy reform has backfired an uncertain tenancy strategy followed and compounded failure to deliver land to tillers. Its shortfalls have made tenants even less secure than previously in their occupancy and conditions of labour. This is because it has been fully possible for landlords to evict tenants and/or put them on different annual terms. In this way, landlords rather than tenants have been liberated from the obligations inherent in feudal relations. Plans to enable registered tenants to gain a share of the tenanted land were sound but weakly implemented. They also provoked a classic ill of failed reforms, a rise in under-used farmland in the midst of land shortage, landlords preferring to leave their lands idle (and being permitted to do so) than to risk tenants claiming partition rights.

Lack of will to reform lies at the root of failures. Reasons for failure in Nepal's land reform mirror those seen in other weak redistributive reforms. The immediate cause has been the repeated leaving of space for landlords to avoid or manipulate new legal requirements (and through this putting rule of law itself in jeopardy). This shortfall in turn stems from weak will to succeed. This has been,

compounded by a too imitative approach, relying upon orthodoxies which were never going to be easy to achieve in the best of circumstances and even less so in the progressively challenging times of the last half century. In many ways, and especially since the 1980s, Nepal was too late with adoption of a reform primarily built around militant and post World War conditions. This has not been an experience unique to Nepal, quite common in Latin America. Successful reforms (and there have been a number, especially the 1940s to 1960s) have avoided this by being characterized by -forceful, comprehensive and speedy implementation, preventing lack of (a) adherence or avoidance by landlords (the most successful were completed within two to five years); substantial support for beneficiaries beyond provision of land, enabling (b) them to launch self-reliant and competitive farming as independent producers; backing up the sector as a whole with significant investment in irrigation (c) and roads, seed and fertilizer supply, technology, and cost-effective marketing systems; recognition of the power of indebtedness to prevent and derail reforms (d) and making credit not just cheap but directly and easily accessible to the very poor, and on a sustained basis; the nesting of land reform in a well-thought through agricultural (e) investment strategy and linking this to intelligent off-farm light industrial development; in particular keeping redistribution out of the marketplace by (f) compensating landlords at well below market values and in shares or bonds which allow them to access their compensation only by investing in off-farm productive enterprise, and with encouraging supporting assistance from the state; and involving beneficiaries from the outset in implementing, regulating and (g) monitoring reforms. None of these conditions have existed in Nepal despite some being recognized as essential, most comprehensively by the High Level Land Commission of 1994-95 and now rearticulated to an extent in 2008. Driving this has been equally characteristic lack of genuine political will, unwillingness to sacrifice privilege, or to fracture longstanding shared interests of the landlord and bureaucratic elite. Ironically, Nepal's emerging democratization over the same period has not helped, depriving the state of the militant autocracy which, for better or worse, has been a common factor in those land reforms which performed well during the 20th century. In recent years the replacement of autocracy with popularly-driven reform has become the logical precondition of successful land reform. This requires however much more devolved forms of land governance to work. While constraints and challenges abound, positive conditions for reform exist. New Nepal has revitalized its commitment to land reform. This is now termed scientific land reform to denote the inseparability of redistribution and growth and to emphasize the investment needed in the farm sector, alongside removing grossest inequities and labor exploitation. Positive conditions for seeing this through exist including -Growing frustration with the continued stagnation of agriculture and a) acknowledgement that failure to reform is a main impediment that must be removed. The reshaping of democracy towards more genuinely inclusive b) representation of landless, land poor and exploited sectors excluded in the past. A more politicised society overall in which the rural poor begin to find c) their voice. The experience of civil war which has demonstrated how real land d) grievance is and how easily grievance can turn into violence. The fact that all parties in principle acknowledge feudal relations must e) be done away with once and for all and almost all have signed up to 'scientific reform' as the vehicle.

Human Resource Management Software

A human resource information system (HRIS) is software that provides a centralized repository of employee master data that the human resource management (HRM) group needs for completing core human resource (core HR) processes.

An HRIS stores, processes and manages employee data, such as names, addresses, national IDs or Social Security numbers, visa or work permit information, and information about dependents. It typically also provides HR functions such as recruiting, applicant tracking, time and attendance management, performance appraisals and benefits administration. It may also feature employee self-service functions, and perhaps even accounting functions.

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In some ways, an HRIS can be considered a smart database of employee information. The interaction of the data, the processes that can be performed and the reporting capabilities make the data stored in the system more accessible and usable.

HRIS Benefits

An HRIS enables the HR department to spend less time on clerical tasks, helps ensure the accuracy of employee data and can enable employees to take a greater role in the management of their information.

Having a centralized repository for employee data removes the need to store paper files, which can be easily damaged, as well as the need to search through large paper-based employee files to find information. Depending on the type of HRIS software, it may generate various reports, provide ad hoc reporting capabilities and offer HR analytics on important metrics such as headcount and turnover. Modern HRIS software also offers visualization capabilities for employee data, such as automatically rendered organizational charts or nine-box grids.

When an HRIS has employee or manager self-service, the process for making employee master data or organizational changes becomes more efficient and uses less time than with paper-based requests. Approval workflows enable changes to be approved or rejected, with the necessary individuals automatically notified. An HRIS might also offer mobile capabilities that extend self-service and provide additional flexibility for remote workers.

HRIS security and privacy

An HRIS also helps secure employee data and keep information private. When using paper forms or spreadsheets, information can easily be accessed by people who may not have the authority to access it. An HRIS can secure information so that it can only be accessed by authorized individuals.

Data security and privacy are important factors when handling sensitive personal information, especially in countries like Germany or France, where works councils have a strong role in protecting employee data. With the exception of lock and key, protecting paper records can be extremely difficult.

Types of HRIS software

A variety of HRIS systems are available and aimed at different types of customers, ranging from small and medium-sized businesses (SMBs) all the way up to large enterprises. Usually, the difference is in the range and depth of features for each process area.

While most HRIS systems cover a large portion of the processes described above, many HRIS systems aimed at small to medium-sized enterprises (SMEs) have less depth of functionality in each feature than those aimed at large enterprises.

In this way, the HRIS market is similar to the automobile market. All automobiles will get a driver from A to B, but major differences exist in the quality and amenities offered.

HRIS functions

As an HR tool, an HRIS usually features modules to handle the following tasks:

- Master data management (MDM)
- Organizational management, such as positions and departments
- Employee and manager self-services
- Absence and leave management
- Benefits administration
- Workflows
- Performance appraisals
- Recruiting and applicant tracking
- Compensation management
- Training tracking (as opposed to a learning management system [LMS]) and organizational development
- Reporting and basic analytics

An HRIS provides a comprehensive set of human resource management functionalities to serve most HR needs. Without this, unsecured or paper-based documents or spreadsheets are required to store data. Manual data entry can cause errors and manual cross-checking of documents and spreadsheets can be

time-consuming and sometimes confusing, especially with a lack of standardization in how data is captured and stored.

Even when a specific system is purchased to cover a process -- such as benefits administration -- it may mean manually entering employee data changes to keep the system up to date. If multiple systems are used, data re-entry may be required for each system, or users may need to export data from one system, change it and then import it into another system.

In some instances, payroll can be part of an HRIS. However, many vendors either don't have payroll as part of their HRIS offering, or -- as with Oracle, Workday and SAP Success Factors -- they sell payroll as a separate system that integrates with their HRIS.

Importance of HRIS

An HRIS can play a critical role in enabling compliance -- for example, to store regulatory data for a country, such as U.S. equal employment opportunity information or U.K. Working Time opt-out -- and can offer a means of gaining insight into the workforce. Both are important and, in some industries, are interwoven.

In addition, downstream integration of systems that require employee data, such as payroll or LMS, and the immense time savings from having integrated applications means an HRIS can serve a critical role, since data entry in multiple systems -- a reality for organizations without an HRIS -- can lead to costly errors or reduced employee engagement.

As one example, suppose a company that manually enters HR data mistakenly overpays employees or gives out too much vacation time. That company will find employee engagement negatively affected if the error is reversed, a situation that could be avoided with an HRIS.

The difference between an HRIS and an HRMS

Exact definitions for HRIS and human resource management system (HRMS) vary, but many experts take the view that an HRMS offers greater functionality by adding talent management and human capital management (HCM) options to human resource information systems.

The talent management functions often include:

- Employee on boarding processes
- Succession planning
- Career development planning
- Learning management

The HCM functions often include:

- labor tracking, typically as a system that tracks all necessary work and distributes that work to workers, often in hourly roles, such as in manufacturing plants;
- time entry and evaluation; and
- workforce management.

HRIS Analysts

HRIS analysts are highly trained professionals with skills in both IT and HR, who are responsible for managing the HRIS and presenting relevant and beneficial data on employee productivity, attendance, training and pay. HRIS analysts also ensure IT departments adhere to HR regulations as well as provide necessary resources to employees and arrange for appropriate equipment updates. Large organizations may employ several HRIS analysts to focus on specific HR tasks, such as employee benefits, compensation or training.

In general, HRIS analysts ensure efficient organization and presentation of information concerning all features of HR functions within a company. Some specific benefits HRIS analysts provide include:

- Customer service for both the employee users of the HRIS and the management users.
- Advice based on analysis of HRIS processes and outcomes from someone who specializes in the program and its performance.
- Data entry for the large amounts of employee information that is gathered.
- Assurance that employee information and data is kept confidential and secure.
- Increased accuracy due to the analyst's editing and confirming of data before it is reported.

At this time, HRIS analysts are not required to possess any sort of certification. However, in order to be competitive in the job market and increase salary potential, it is suggested that applicants provide proof of their excellence in the field and commitment to HR by obtaining certifications such as the Professional in Human Resources (PHR) or Senior Professional in HR (SPHR) certifications -- both from the Human Resources Certification Institute ([HRCI](#)) -- and the Human Resources Information Professionals (HRIP) certification from the International Association for Human Resource Information Management ([IHRIM](#)).

INDIA

NICNET (NICNET-Role of Nationwide Networking in E-Governance)

Introduction

National Informatics Centre (NIC) is a premiere S&T institution of the Government of India, established in 1976, for providing e-Government/e-Governance Solutions adopting best practices, integrated services and global solutions in Government Sector.

In 1975, the Government of India strategically decided to take effective steps for the development of information systems and utilization of information resources and also for introducing computer based decision support system (informatics-led development) in government ministries and departments to facilitate planning and program implementation to further the growth of economic and social development. Following this, the Central Government nucleated a high priority plan project "National Informatics Centre (NIC)" in 1976, and later on with the financial assistance of the United Nations Development Program (UNDP) to the tune of US\$4.4 million.

When Government of India approved NICNET, the National Information center had to evaluate various technology options of the day and come to an optimal solution for a nationwide data network. In view of our national success stories in satellite communication, be it in broadcasting, television or voice, a satellite based solution for the purpose of linking all the state capital and district headquarters through a data network appeared most appropriate. It gave a solution which was distance-independent, cheap, scalable and easily deployable. Once the satellite option was made, the issue of mechanism of channel sharing was addressed. To provide two way communication to a large number of nodes using classical time Division multiple Access (TDMA)/Frequency Time Division Multiple Access (FTDMA) mechanisms of carrier sharing were not feasible unless costlier and complex sharing mechanisms were incorporated.

Hence it was decided to go for a Code Division multiple Access (CDMA) based on Very Small Aperture Terminal (VAST) networking under the NICNET program. Commercially viable, cheap VSATs in C-band using CDMA of nodes for the same carrier in the two-way data link. The advantages of the CDMA. VSAT option chosen were specifically

- Small size (1.2 dia)
- Low cost
- Ease of installation and deployment
- Easy scalability

The VSAT had a 1200 bps uplink and 19.2 kbps downlink, through a few VSAT's had an uplink of 9600 bps. This low speed was, of course, a limitation.

Using VSATs as communication link some of the pioneering applications of NICNET in government were the following:

1. **Electronic Mail for Government Communications:** NICNET provided pre-internet, electronic mail to all secretariats of state governments, ministries of Government of India, and district collectorate through a centralized e-mail service.
2. **Database Access:** NICNET helped its users to access database like GISTNIC (General Information Services Terminal-National Informatics Centre), MEDLARS (Medical Literature Analysis and Retrieval Systems), etc. from remote locations.
3. **Talent:** NICNET enabled remote login of systems across the country.
4. **File Transfer Protocol (FTP):** NICNET enable files transfer across states and districts. This application was widely used for, applications like budget transmission and election result analysis.

History

The NIC was established in 1976 under the aegis of the Ministry of Electronics and Information Technology.[8][9] The NIC is credited with helping the Indian government embrace IT in the 1990s[10] and has also helped disseminate e-governance to the masses.[11]It had an annual budget of ₹11.5 billion (US\$160 million) for the year 2018–19.[2]

In May 2019, the government of India set up the Centre for Smart Governance (CSG), and state governments have since been advised to consult the CSG for IT projects they previously would have consulted the NIC and private firms for. Some claim that government sources have said "NIC is said to be unable to scale up", and Rajeev Chawla, Additional Chief Secretary (e-Governance), was quoted as saying "CSG will be an analogue to NIC"

Achievements

NIC has emerged as a “prime builder” of e-government/e-Governance applications in government sector (national, state and local districts) as well as promoter of digital opportunities for sustainable development, during more than a quarter century period. NIC has institutional linkages through its ICT Network “NICNET”, with all the Departments/Ministries of the Central Government, 28 State Governments, 1 National Capital Territory of Delhi, 6 Union Territories, and about 600 District administrations of India. NIC has been instrumental in steering e-Government/e-Governance applications in Government Ministries/Departments at Centre, States, District and Blocks, facilitating improvement in government services, wider transparency, promoting decentralized planning and management, resulting in better efficiency and accountability to people.

NIC has been involved implementing “eGovernance agenda” of the Central Government with respect to:

- Internet/Intranet Infrastructure (PCs, Office productivity tools, Portals on Business allocation) upto Section officers levels;
- IT empowerment of Officers/Officials & Capacity Building
- ICT Enabled Services (G2G, G2E, G2C and G2B)
- ICT Plans for Sectoral Informatics Development;
- Business Process Re-Engineering
- Services profiles, among the others, include:
- Network services (WAN, MAN, LAN)
- Capacity Building through Human Resources Development of Government Employees;
- Datamining and data warehousing
- Total ICT Solutions
- Video Conferencing & web services
- Certification Authority and PKI Services
- Domain (gov.in) Registrar
- Computer Emergency Response Team (CERT) Services
- National Disaster Recovery Centre
- Geomatics & Informatics design and development for decision support

- Sectoral ICT Plan formulation
- ICT projects consultancy

Strengths of NIC

- Domain expertise in various sectors of Government Business Allocation (Social Sectors, Economic Sectors, and Accounting and Treasuries, etc);
- Development Expertise & Experience in Systems Development Life Cycle (SDLC);
- Expertise & Experience in Networking, Software Technology and Hardware technology;
- Web sites development and hosting with expertise in developing dynamic sites;
- Email and Internet services using NICNET;
- Imparting training in 'standard tools', computer awareness and application systems;
- Handholding support during implementation;
- District Centers providing state-wide and nation-wide support for application systems;
- "NIC is, perhaps, the only S&T organization which has the infrastructures to perform the functions of four different service providers envisaged in the Convergence Communication Bill 2000, namely:
 - Network Infrastructure Facility Provider (NIFP)
 - Network Services Provider (NSP)
 - Application Service Provider (ASP)
 - Content ASP"

Conclusion

In this case study we have studied about history of NICNET and how VSAT was established and its application. NICNET was upgraded to meet the present day requirements of the governmental various levels. Also we studied about achievement and strength of NICNET.

Collectorate

Introduction

Collectorate 2000 is a program of the National Informatics Centre(NIC) to design, develop and install new suite of software modules for use is the computerization of Collectorates. This case study surveys the background, features and capabilities of the software modules used in the computerization of Collectorates in Andhra Pradesh.

During the late 1980s, NIC took up the NICNET program for implementing the All India Satellite based Government Informatics Network connecting various District Collectorates, State Government Secretariate, and Central Government Offices in Delhi through a VSAT based network called NICNET.

In 1988, VSATs have been installed (C200 model) in all District Collectorates with a speed of 1.2 kbps along with one computer system and two Technical Officers of NIC. Subsequently, in mid 1990s, the NICNET facility was enhanced with IPA/FTDMA VSAT Earth Station with a speed of 64 kbps (shared) for Internet browsing purpose in the Collectorate.

Collectorate 2000 At District Collectorates

Project "Collectorate 2000" (C2K) was conceived by National Informatics Centre (AP State Unit) during April/May 2000. This project, as explained below, covers all the functional processes in the District Collectorate in a comprehensive manner.

The District Collectorate is the seat of the State Government in the District. All the State Government Departments are represented in the District at the Collectorate and associated offices.

Functions of the Collectorate:

- encompassing the revenue functions,
- land related matters and the magisterial
- functions dealing with Law and Order, Arms Licenses, etc.,
- Welfare schemes, relief and rehabilitation

The Collectorate 2000 software developed by National Informatics Centre (AP State Unit), Government of India, Ministry of Information Technology, attempts at automating all the above identified functions with about 24 software modules, each module relating to individual function.

Software Modules in Collectorate 2000 Software

Table: List of Software Module in Collectorate 2000 Software

	Subject/Department		Module
A.	Land	1.	Land Alienation
		2.	Land Assignment
		3.	Land Encroachment
		4.	Land Transfer
		5.	Land Acquisition
		6.	Land Lease
B.	Revenue	7.	Land Revenue (DCB)
		8.	NALA
		9.	Revenue Recovery
		10.	Crop Damage
		11.	Court Cases-Collectorate

C.	Magisterial Functions	12.	Gun Licenses
		13.	Gun Licenses
D.	Directorates	14.	Food and Civil Supplies
		15.	Family Welfare
		16.	NSS
E.	Welfare Schemes	17.	Disabled Welfare
		18.	Apatbandhu
		19.	Gruharaksha
		20.	Drinking Water
		21.	Freedom Fighter Pension
F.	Petitions	22.	Petition Monitoring (Web based)
G.	Govt. Orders	23.	G.Os Databse (Web based)
H.	Geographic Information System (GIS)	24.	GIS for District Level

Modules of Collectorate 2000

The following is a brief description of each of the major modules of Collectorate 2000;

1. Land Alienation and Land Transfer Module

Allotment of Government land to Government/Private Organizations for public purpose is known as Land Alienation. In case it is allotted to State Government department/undertakings, State Government undertakings, local bodies, or private organizations. The alienation (i.e. allotment) can be on payment of market value by the applicant or for free of cost.

Scope of the Module

This module covers the various stages in the processing of an application for land alienation, i.e. from registering the request details to approving the case and storing the remittance particulars. Various actions performed by the officials at different levels involved in the process are recorded into the system.

Features of the System

- Security provided through user login.
- Privileges of action based on designation.
- Integration with a separate product which handles File Monitoring.
- Can be operated in standalone mode or in association with File Monitoring System.

2. Land Assignment

Allotment of government land to individuals belonging to certain categories like landless poor or ex-servicemen, freedom fighters/sufferers, etc. for housing or agriculture purpose is known as Land Assignment.

Scope of the module

a. Under beneficiary details

- Beneficiary name.
- Address of the beneficiary
- SSID (Social Security Identification No.)
- Father/Husband name of the beneficiary
- SSID No. of father/husband
- Sex, Age, Caste
- Category (Landless poor/Ex-servicemen/Freedom fighter etc.)
- Annual income
- Type of land assigned
- Location details of assignment like
- Village, Survey No., Extent of land assigned
- Assignment date

b. Under Monthly Progress reporters: The data on number of beneficiaries persons affected and the extent of area on the following items:

- Illegal assignments
- Sivai Jamadar Occupied and regularized
- Illegal Sivai Jamadars Occupied and evicted
- Encroachments reported
- Cancellation order issued
- Evictions made

Output

- a. Beneficiaries lists (Mandal/Sub-division
- b. Consolidated Report (District-Total) for Statement-I, Statement – II, Statement–III.,

3. Land Revenue

The objectives is to consolidate Agricultural Land revenue and Non-agricultural Land Assessment (NALA) at the District Collectorate form the Mandals for monitoring the collection of Revenue. NALA is the revenue from the residential, commercial and industrial establishments.

Both the Agricultural Land Revenue and the Non Agricultural Land Revenue include statements of the Demand Collection and Balance (DCB) and the Non Collectable Balance (NCB). The Statement of DCB is consolidated daily/weekly/monthly. Non-collectable balance is consolidated monthly.

The district-wise consolidated statements are sent to the Chief Commissioner of Land Administration.

Input

DCB, Demand, Collection and Balance of Agricultural Land Revenue. This is captured daily/weekly/monthly.

NCB. Monthly statement of Non Collectable Balance of Agriculture Land.

NALA-DCB. Demand, Collection and Balance of Non Agricultural Assessment. This is captured daily/weekly/monthly.

NCB. Monthly statement of Non- collectable Balance of Non Agricultural Assessment.

Output

- a. District level consolidated daily/weekly/monthly reports of DCB of Agricultural Land Revenue to be sent to Chief commissioner of land administration.
- b. District level consolidated monthly reports of Non Collectable Balance of Agricultural Land Revenue.
- c. District level consolidated daily/weekly/monthly reports of DCB of Non Agricultural Land Assessment (NALA).
- d. District level consolidated monthly reports of Non Collectable Balance of NALA.

4. Food and Civil Supplies

Food and Civil Supplies module allots essential commodities like rice, wheat, sugar, kerosene oil, etc. to fair price shops based on a criteria and card strength in a shop. Dealers Certificates and Photo Identify Cards are generated using the scanned images of photographs of the dealers and monitors the stock lifting and Fair Price Shops performance.

The major modules of the system are:

- Dealer's information, ration card and stock backlog details
- Allotment of essential commodities
- Lifting of essential commodities.
- Inspection of the fair price shops

- District-wise/mandal-wise/shop-wise reports to cater to the needs of users.

5. **Gruharaksha**

Gruharaksha is an insurance scheme for Below Poverty Line (BPL) families to cover the loss or damage to their dwellings due to natural calamities. This module computerizes the complete procedure starting from accepting claimants applications to disbursement of amount and its reimbursement from insurance agency.

The major models of the system are:

- Acceptance of claimants details and its scrutiny
- Calculating eligibility amount monitors the sanction/budgets
- Storing the dispatch details
- Reimbursement of money from insurance firm.

6. **Apathbandhu**

Apathbandhu scheme is a social security scheme launched by Government of Andhra Pradesh. If a bread earner dies due to accident, the scheme provides assistance to legal heirs of bread earner. The scheme is applicable to Below Poverty Line (BPL) families. The scheme is launched in collaboration with New India Assurance Company Ltd. (NIAC). The software designed automates the complete scheme starting from allocation of budget to districts upto the sanction of assistance to claimants.

Major components of the module are:

- Maintaining the budget.
- Acceptance of the claim from received from the Mandal Revenue Officers.
- Sanction of assistance
- Generating the statutory claim forms as prescribed by NIAC-both for the road accidents and other than road accidents.
- Automating the budget availability based on the reimbursements received from NIAC.

The scheme covered are

- Family Welfare (FW)
- Reproductive Child Health (RCH)

7. **Law and Order Information System**

This module is an excellent monitoring and decision support tool to the collectorate, In all the collectorates the matters concerning law and order are dealt by seat C1 under section C. Various issue dealt with are:

- i. Extremists and naxalities
 - Surrendered extremists

- Civilians killed /injured by extremists
- Property damaged by extremists
- ii. Deaths occurring in
 - Lockups
 - Communal violences
 - Open Firing
 - Accidents

8. Drinking Water Program Monitoring System

This package deals with works monitoring during adverse seasonal conditions in the mandals within a district. The details on fund estimation, release and utilization with respect to various schemes/works are maintained and monitored in the system.

Features of the System

- Entry of normal and the actual rainfall details and cost estimates received at the Collectorate from the executing agencies on the works to be taken up.
- Release of respective amounts to the executing agencies
- Capture of details on different stages of work like administrative, technical sanctions and grounding, completion of work Generation of mandal abstract on schemes of works taken up by different executing agencies with details on fund estimates, releases and expenditure.
- Provision of a template with District/Mandal/Villages, inancila year, Scheme/Work and Estimates/Release /Expenditure/Concsolidated options for generating various free format reports.

Computer-Aided Administration of Registration Department (CARD)

Introduction

CARD (Computer-aided Administration of Registration Department) project was originally conceived in August, 1996 and implemented at two test sites at Registrar Office, Hyderabad and Sub Registrar Office, Banjara Hills in August/September, 1997. Judging the project by its initial success and the immense potential it has in transforming the concept of public service, the Government decided to replicate it all over the State.

The CARD project aimed at the complete computerization of the land registration process in AP. The case highlights the problems faced by the citizens of AP before the implementation of CARD. The CARD Project provides a transparent method of valuation of properties and calculation of stamp duties, simplified the registration procedures, enhanced speed, reliability and consistency of the system. It reduced delays by replacing the manual systems of copying, indexing and accounting. Retrieval of the documents and obtaining copies is made instantaneous. The overall effect is a smooth public interface. A highly complex but user friendly CARD Software was developed by the team of National Informatics

Centre, Hyderabad. The required hardware for implementation of CARD Project has been procured through A.P. Technology Services Limited, Hyderabad.

History

214 Sub Registrar Offices in respect of A,B,C category offices were inaugurated on 4-11-1998 and legal status was accorded to the CARD Project through “Act 16 of 1999” w.e.f.5-2-1999. After successful implementation of project in the above sites another 25 “D” category offices were implemented with the CARD Project and legal status was given w.e.f.1-11-2001. Finally the remaining 148 E& F category offices (each 74) were computerized and the legal status was accorded from 10-3-2002.

Present Scenario

The CARD system had been in the client server architecture with each Sub-Registrar Office having one server with standalone functioning. Later all the Sub-Registrar Offices were provided with network connection for obtaining the index data and MIS data to the central server. In the year 2011 Government have taken a decision to bring CARD into Centralized Architecture called CCA (CARD centralized architecture).

CARD was initiated to meet the following key objectives:

- Demystify the registration process
- Bring speed, efficiency, consistency and reliability
- substantially improve the citizen interface

These goals were to be achieved by:

- Introducing a transparent system of valuation of properties, easily accessible to citizens.
- Replacing the manual system of copying and filing of documents with a sophisticated document management system using imaging technology
- Replacing the manual system of indexing, accounting and reporting through the introduction of electronic document writing.

Benefits and Costs

Six months following the launch of the CARD project, about 80% of all land registration transactions in AP were carried out electronically.

The time required for services such as valuation of property and provision of certified copies of registered documents now takes 10 minutes instead of a few days as under the earlier system. ECs are now issued to citizens in a span of 5 minutes, using a system that searches through more than 15 years of records from over 50 offices. Land registration can be completed in a few hours, whereas earlier it took 7-15 days.

After factoring out the natural upward trend in nominal revenues, the CARD system has generated a modest increase in revenue. The growth rate in net revenue collected has gone up by almost 20%.

The main advantage of this system has been the awareness in rural public towards the benefits of ICT in important activities and so the public is demanding for more such projects at the rural level.

Procedure for Registration of Documents

The registration system is governed by antiquated procedure which include the laborious copying and indexing of documents as well as the unscientific space consuming preservation in ill-maintained back rooms (record rooms).

The tedious procedure and lack of transparency in property valuation resulted in the flourishing business of brokers and middleman who exploited citizens selling/buying properties.

The conventional 13-step registration procedure is complex and time consuming and beyond the comprehension of most of the citizen. It involves

- Step 1:** Ascertaining the market value of the property
- Step 2:** Calculating stamp duty, transfer duty and registration fee
- Step 3:** Purchase of non-judicial stamp paper
- Step 4:** Getting the legal document prepared
- Step 5:** Procuring the necessary certificate and enclosure to be enclosed with the document, as for example, no objection certificate from ULC (Urban Land Clearance) and PTA (Paths Transfer Application)
- Step 6:** Presentation of the document to the SRO of the jurisdictions
- Step 7:** Security of the documents by the sub-registrar (SR)
- Step 8:** Valuation of the property by the SR, Calculation of the stamp duty (SD), transfer duty (TD) and registration fee (RF)
- Step 9:** Payment of deficit stamp duty if any
- Step 10:** Admission of execution by the execution before the sub-registrar and two witnesses
- Step 11:** Copying of the documents in the register books
- Step 12:** Posting of entries to two indexes (by name and property) accounts and reports
- Step 13:** Return of the document to the citizen.

The following is a brief account of the various service agencies involved in the above 13-step process.

- a. Stamp vendors:** 221 department counters and 300 STOs (Ex- officio stamp vendors), 2300 licenced stamp vendors
- b. Document writer:** Licenced 3900 Documents writer: Now any one can prepare a document without any additional payment of fee
- c. Registration agents:** Self employed individual or firms
- d. Registrations offices:** 387 Sub-Registrar offices

28 Registrar offices

6 Deputy Inspectors General Offices

1 Inspector General Office

Limitations

As CARD system primarily deals with registration of immovable properties. It is essential to have a well-defined standard for describing the various types of properties. The absence of such of standard will effect the process of valuation search for encumbrance, detection of prohibited properties such as government lands, assigned lands, and surplus lands. Introducing the standards in property description duly taking care of the following can further enhance the dependability and reliability of the system.

- a. Natural changes in the classification of properties over a period of time, i.e. from Agriculture-Non-Agriculture. Plot-Houses/Apartments/Multi-storeyed building.
- b. partitions/Merging of properties.
- c. Different nature of right like Undivided share, Joint/Common rights and intersects on properties.
- d. Lack of well defined (unique) identification (ID) rules for the properties, particularly for open plots, coupled with non-standard id's for house number and survey numbers, etc.
- e. Inconsistent use of Ward and Bloc numbers of Local body offices with that of the other department like Census/Revenue etc.

Smart Nagarpalika Computerization Of Urban Local Bodies (Municipalities)

Introduction

The urban local bodies in the towns and cities of the country are autonomous and are managed by elected representatives. These bodies, also known as Municipalities, are responsible for the management and upkeep of the civic amenities in the town or the city. For this purpose they collect taxes such as property tax, water tax, etc. which form the revenue for the local body/municipality. In addition, the local body is also responsible for registration of births and deaths, and issue of building permissions and trade license. Following is the comprehensive list of functions executed by municipality/local body:

- Birth and deaths registrations
- Property tax
- Water tap connections and charges
- Advertisement tax
- Financial accounting
- Building permissions
- Grievance redressal

- Project and works monitoring system (PROMS)
- Dangerous and offensive trade license
- Assets and inventory
- Court cases monitoring

The NIC, Under the Department of Information Technology of the Ministry of Communications and Information's Technology, Government of India, has taken the lead in constituting a national level project called "SMART Nagarpalika" wherein all the above subjects have been studied in-depth with Andhra Pradesh as the pilot State. A detailed systems requirements specifications (SRS) report has been prepared and the same has been utilized as basis for developing the software on the above subjects. The following sections deal in details with the functionalities of each of these software modules:

Births and Deaths Registrations

Registration of Births and Deaths is one of the major activity of Municipality and is done in compliance with the AP registrations of Births and Deaths rules 1999, which are drawn from the Registration of Birth and Deaths Centre Act 1969. Registrar-General appointed by Central or state government maintains the registers of births and deaths recorded.

Birth and death occurring within the geographical zone of a municipality has to be reported to that municipality within 14 days in case of births and 7 days in case of deaths. Hospitals are responsible for reporting the event of a birth or death occurred at their premises; if the event occurs at home then head of the family or any neighbor is responsible for reporting.

For the birth or death which is not reported within the specified period, but reported within one month of its occurrence, late fee is applicable. In case birth and death reported after one month and within one year of its occurrence, late fee and affidavit have to be submitted. If the reporting is still delayed, then magisterial order and non availability certificate have to be submitted along with the late fee.

Objectives

The following are the objectives of the births and deaths registration module

- Registering of births and deaths
- Issuing of birth, death and non-availability certificates
- Generation of field verification report
- Generation of birth and death registers
- Generation of monthly and yearly statistical report
- Closure of the registrars at the end of year
- Provision for inclusion of name.

Salient Features

The Following are the salient features of births and deaths registration module:

- Maintaining the fee collections
- Standardization of master details
- Maintenance of statistical information
- printing of acknowledgement/receipt
- Online status enquiry

Property Tax

Property tax is a tax paid on property owned by an individual or other legal entity, such as a corporation. Most commonly, property tax is a real estate ad-valorem tax, which can be considered a regressive tax. It is calculated by a local government where the property is located and paid by the owner of the property. Property tax is one of the prime revenue sources of the municipalities. Property tax is levied by all municipalities by a resolution of the Municipal Council. The tax comprises components such as tax for general purpose, water and drainage, lighting and scavenging. Property tax aspect is handled by revenue section in the municipality. The various functions concerning to property tax are assessment of buildings/lands, current/arrear-demand collection, title transfers, revision petitions, appeal petitions etc.

Objectives

- Standardization of procedures and norms for assessment and collection leading to uniformity.
- Standardization of revenue zones, wards, blocks, locality, complex apartments, building age and depreciation.
- Ease in tracking demand and collection of assessments category-wise.
- Fixation of monthly rental values, tax standards, penalty rate and tax components.

Water Tap Connections and Charges

The Municipalities on request issue water tap connections to the owners of the buildings. Connections are sanctioned based on the category and availability of resources by the municipal commissioner. Connection charge and deposits are to be paid by the citizen. Water tap connections are of four types: domestic non-metered tap, domestic metered tap, commercial metered tap, industrial metered tap. The water tap connection may be disconnected by the Municipality in case of long standing dues from the user. The re-connection is possible by clearing the dues and paying the necessary re-connection charges. Illegal tap connections can be regularized by paying regulations charges.

The various functions concerning water tap connections and charge are issuing water tap connections, regulations of illegal tap connections, work order, demand/arrear raising and collection of tariff, etc., all achieve the following objective:

Financial Accounting

The accounting system being followed in municipality flows from a system adopted in 1930s where the local bodies have control over all the activities being undertaken in their domain. The tax structure is also

designed to meet the expenditure incurred in the fields of lighting, public, health, preventive medicine, scavenging, education-elementary and secondary, water supply and drainage, town planning, maintenance of remunerative enterprise like markets, slaughter houses, etc. and endowments. The accounting system is so designed that income and expenditure in each of the areas can be ascertained at the end of the financial year and tax structure can be suitable modified. The general fund is meant for creation of new assets and improvements of facilities in the area under the municipality, Based on the activities in municipality, the functions are assigned to various section for better monitoring and control. The major sections in the municipality are revenue, engineering, town planning, public health, accounting etc. The financial accounting module being developed will address the requirements of these sections by integrating the activities.

Assets and Inventory

The web enabled Assets and Inventory module establishes assets and inventory management for the municipal administration. The asset will keep track of complete information of assets owned by the municipality right from age old till today. The inventory part will keep record of the items procedure for the maintenance of the assets and serviceable goods.

The functional requirements are, recording right from the nature of the asset as and when it arises, its maintenance procedures and event of asset sale. The inventory of the materials used for the maintenance of the asset and serviceable goods such as medicines, engineering items, stationary, which are classified as consumable, are recorded. The stock position of the materials dealt in the municipality is taken care of.

Conclusion

This case study surveys the software for computerization of Urban Local Bodies (Municipalities) under the title "SMART nagarpalika-Computerization of Urban Local Bodies (Municipalities)". Various subject such as Births and Deaths, Property Tax, Water Tap connections and Charges, Advertisement Tax, Financial Accounting, etc. are covered in details in terms of their software functionality and capabilities. This software is being implemented in many Municipalities in the country. It is also being supported by the Ministry of Urban Development.

National Reservoir Level and Capacity Monitoring System

Water is a prime natural resource, a basic human and a precious national asset, and hence planning and development of water resources need to be governed by national perspectives. The National Water Policy (1987) has addressed the issue related to develop, conserve, utilize and manage this important natural resource in this new millennium. The Hashim Committee Report (1999) dealt with "Integrated Water Resources Development Plan" warrants timely action in respect of (i) harnessing of the surface flows through major, medium and minor storage, (ii) improving water-use efficiency to the optimum levels, and (iii) taking necessary steps for demand and supply management. Emergence of Information Technology on the national agenda and the announcement of IT policies by various State Governments have recognized the "Convergence of Core Technologies and E-Governance" as a tool for sustainable development in water resources.

Information and Communication Technology (ICT) applications in water resources development and management in India are being given adequate attention during the Tenth Plan, facilitating deployment of

Decision Support System (DSS) of various levels of planning and management of water resources of the country. Evolved in this direction is "Reservoir Level and Storage Capacity Monitoring" – A segment of e-Governance in water resources. This is a network-centric application and has a significant importance in evaluation of water utilization, crop production and estimation, hydropower generation, of states/national level under uniform protocol of communication among stage and centre. The parameter could be an early warning indicator to the crop production and hydel power generation of the country. Any significant deviation from the norms could attract attention of the planners and administrators.

National Information Centre (NIC) has been providing support since 1988 to CWC for monitoring 70 reservoirs with the total storage capacity of 135 TMCM comprising 78% of the total available storage, supporting CCA of the order of 11,534 Th. Ha., hydel power capacity of 8617 MW and covering the reservoir, spreads in more than 14 states and 12 major river basins.

National/State Model

With the current available technology of Internet and GUI environment, it has become possible under Java and SQL Server environment (Web-enabled application) to evolve a new version State/National Model to provide support to the State Government, CWC, MOWR and Planning Commission with complete distributed administrative and management control. The information thus collected from the various States can be published (with the concurrence of officials concerned) on the Internet for general viewing of the various departments of Central and State Governments, the Planning Commission and the concerned farmers (Clients), within the prevailing restricted structure of communication.

Important Features

Let us now review the important features of this model.

Users and Privileges. Only authorized user can assess the system. The users of the system have been classified into three categories as follows:

- (i) **Super user [CWC user]:** Has privilege to access all the reservoirs in India and also have permission to access Admin module for creating user and creating reservoir, i.e. complete administrative control.
- (ii) **State user:** Has privilege to access all the reservoirs of their particular state and also permission to create and modify user of reserve level user.
- (iii) **Reservoir Level User:** Has a right to access the particular reservoir for data entry, query and reports of concerned reservoir.

Login procedure

The reservoir level and capacity monitoring system could be assessed through a browser to access the "National User Interface".

National Level User Interface

The application software recognize each of the users allocated, its privileges and security aspects, and hence provides access at National/CWC level, State level, and at Reservoir level, and projects the

respective interface of application area to the concerned user. The format of the "National Level User Interface" is same as the Super User (CWC) shown above.

- i. **Data Entry:** User can feed the data on the basis of their privileges
 - Super user can feed data of any reservoir in India.
 - State user can feed data of all reservoirs in the particular state.
 - Reservoir user can feed data of a particular reservoir.
- ii. **Query.** Different kinds of queries for National/State Level as per their privilege include
 - Data entry of a particular reservoir
 - Data of a particular reservoir on the basis of data
 - Particular data of all reservoirs.
 - Query about users' privileges
 - General query about salient features of a particular reservoir.
- iii. **Administration.** The module provides complete administrative control at the national level as well as the state level, for their respective regions as per the privileges defined. Following are the options available:
 - Add user
 - Delete/Update User
 - Add Reservoir
 - Delete /Update Reservoir
 - Add Basin Information
 - Delete/Update Basin Information
- iv. **Reports.** Reports are available under uniform protocol of communication and reporting under NICNET/Internet, for Basin/National State level as per their privileges, reflecting, current year/preceding year and last 10 year average performance.
- v. **Graphical Report.** Capacity versus date graph in near real-time environment for stage/national level as per their privileges.

Conclusion

In this case study we have seen how a critical resources as water level (in various rivers in the country) can be monitored, with the help of a specialized software, at various levels.

Computerization in Andhra Pradesh

Introduction

The operations of the Andhra Pradesh State Trading Corporation (APSTC) have been computerized by National Informatics Centre (NIC), Andhra Pradesh State Unit, Hyderabad. This activity comprises hardware procurement, installation, network installation in the campus of APSTC, systems study and analysis, systems design, software development and implementation, and training handholding of the users. This project was successfully completed and is presently in operation. In the following sections of this case study, the background of the existing system is presented followed by the details of all computerized systems.

The Existing System

Andhra Pradesh State Trading Corporation Limited (APSTC) is an approved terminal operator of Cargo Air Terminal Complex Hyderabad. In this capacity APSTC is authorized to handle the work relating to export, import and unaccompanied cargo.

APSTC cargo terminal complex has four major sections, viz.

- Import godown
- Export godown
- Unaccompanied Baggage (UB) godown
- Accounts section

It has two primary functions, viz. :

- As custodian of import cargo, commercial and incoming unaccompanied baggage.
- Handling of the export cargo as well as outgoing unaccompanied baggage (personal effects).

Summary of the work flow of the cargo operations follows:

Import of Commercial Goods/Unaccompanied Baggage

This involves the following steps:

1. Unloading of cargo/unaccompanied baggage by airlines at Hyderabad Airport.
2. Transportation under customs escort and deposit of import cargo at import godown of APSTC by airlines.
3. Verification of goods against IGM/air way bill by APSTC.
4. Presentation of cargo for examination/inspection to customs by APSTC on requisition by customs.
5. Customs clearance of import cargo on the basis of documents prepared by Importer/CHA (Custom house agent).
6. Raising of bills by APSTC for handling and other service.

7. Collection of APSTC charges from importer/consistence.
8. Preparation of gate pass and delivery of goods to importer by APSTC.

Export of Commercial Goods/Unaccompanied Baggage

The entails the following steps:

1. Exporter deposits commercial cargo/unaccompanied baggage at export godown.
2. Verification of goods by APSTC based on documents presented by exporter.
3. Presentation of cargo for customs inspection/examination by APSTC on customs requisition.
4. Customs clearance of export cargo on the basis of documents presented by exporter.
5. Raising of bills by APSTC for handling and other service, and payment of the same by the exporter.
6. Handling over of the export cargo to airlines under customs escort.

Import of Cargo

APSTC will act as custodian for commercial cargo/unaccompanied baggage arriving at Hyderabad airport. Airport Authorities unload the cargo at Hyderabad airport and the same will be transported in an airlines vehicles under the customs escort to APSTC import godown.

The godown official receives the goods and verifies the same against IGM/Airway bills. Information such as IGM No. address of consignee, number of packets, their weight etc. are entered into a register (Mother Register) and an acknowledgement is issued to airlines. The goods are stacked in godown and their location is recorded in the location register.

The importer files a bill of entry enclosing all required documents with customs official. After verifying the documents the customs authorities will issue a requisition slip for inspection/examination of import cargo. After the requisition by customs official the goods are presented for examination/inspection by godown official. After checkout order, i.e. clearance by customs, the Accounts Section of APSTC, raises a challan/bill for handling storage, demurrage (if applicable), and for other service.

The importer pays APSTC charges and presents the receipt for delivery of goods to godown section. The godown officials prepare the gate pass and deliver the goods to importer. These official also update the relevant columns like delivery order No. and payment details in the mother register before they prepare the gate pass.

The procedure is same by and large for import of unaccompanied baggage (UB).

Export of Cargo

Exporter or his agent deposits the cargo with export godown of APSTC terminal cargo office along with AR4/invoice details. The concerned manager verifies the details against the application and invoice. The details such as consignee's address, number of packets, their weight. etc. are entered in deposit register. He also prepares an acknowledgement and issues to the party/agent.

The exporter will then file invoice details along with other documents with customs. After scrutinizing the document, the customs officer prepares a requisition slip for examination of goods. The APSTC will present the export cargo for examination.

After examination of goods, customs issue a let export order and goods are ready for shipment. Airway bill number is marked on all export packages by godown officials. The accounts sections of APSTC raise a challan/bill for service such as handling, wharfage, forklift, storage etc.

The exporter or his agent pays APSTC service charges and submits LEO, airway bill and payment receipt to export officer for getting a gate pass. The concerned godown official updates the deposit register with challan number, amount paid, etc, and prepares a gate pass for cargo.

Computerized System

Introduction

The computerized system takes care of the existing systems data flow and procedure completely and generates all reports of manual system. It is built around client- server architecture with Visual Basic as front end and MY-SQL server as back end. As computerized system is implemented over LAN, the required data flows automatically from section without manual intervention. Each section gets the screen required for his work of data entry/ updating. It also gets relevant detail for information but he is not permitted to update the information entered in other section.

Security

The following additional precautions were taken to enforce additional security:

1. Users name of person who entered the data/modifications (only one should be maintained by the system.)
2. Deleting the data not be allowed
3. Modification of data by Managers only.
4. Retention of old challan no. in case of the can of the cancellation/modification of challan.
5. In case the payment for the APSTC service charges is made by a challan in bank the godown section will enter the details of the payment into the computerized system.
6. The software should have two independent modules to take care of the old/new charging policies of STC.

Ekal Seva Kendra

Overview

"Keal Seva Kendra" is a progressive step towards people friendly, need based a governance. This has been set up at the district secretariat, Kaithal (Haryana) with a mission of providing time-bounded hassle-free, single –point delivery of service in a professional and citizen friendly environment. Towards achieving this mission, separate counters have been opened with a seven-node Local Area Network in and around the district.

The service under Ekal Sewa Kendra can be summarized as follow:

- Driving licence (Learner/Permanent/Renewal/Duplicate/Addition of Class, etc.)
- Conduct licence (Learner/Permanent/Renewal/Duplicate, etc.)
- Vehicle registration (New/Transfer/NOC/Addition and removal of HPA, etc.)
- Caste certificate
- Residence certificate
- Nakal (Revenue document) service.
- Birth and death certificate.
- Passport application acceptance service.

Apart from these, service from distribution of application forms to medical examination/photography of applicant and delivery of certificates/documents are also provided at these counters. Receipts are being provided to public on which time of delivery is mentioned. In fact, it is a total integrated solution for all these services.

The system is technologically fool proof and upgradable to provide more such services and information. It is a people friendly, need based, technologically empowered and self-sustaining system. Figure C7.1 show service being provided at a counter.

Here are the objectives:

- Time bound delivery
- On the spot computerized scrutiny of documents.
- Single-point contact-service under one roof.
- Simplified procedure-Total transparency
- Easy and effective monitoring-Fixed accountability
- Total solution-All activities computerized
- Elimination of issuance of bogus/fake Licences
- Fully tamper-proof databases
- Middleman is out
- Automatic generation of periodic Account/Audit statements
- Quick search facilities
- Reduced clerical workload
- Online/office line MIS reports
- Self-sustained project.

Technology behind the Kendra

National Information Centre developed and implemented all software used at "EKal Seva Kendra". These software are highly versatile, robust and scalable. it uses Visual Basic as front-end and SQL Server as back-end, based on network topology. Databases are designed, considering the best and latest security techniques.

One server, seven nodes, four dot matrix printers, two laser jet printers, one inkjet printer, digital camera and signature pad are ultimately planned to be installed at the site using local area networking having connectivity with NIC, DC, ADC, SDM Office (See Figure C7.2)

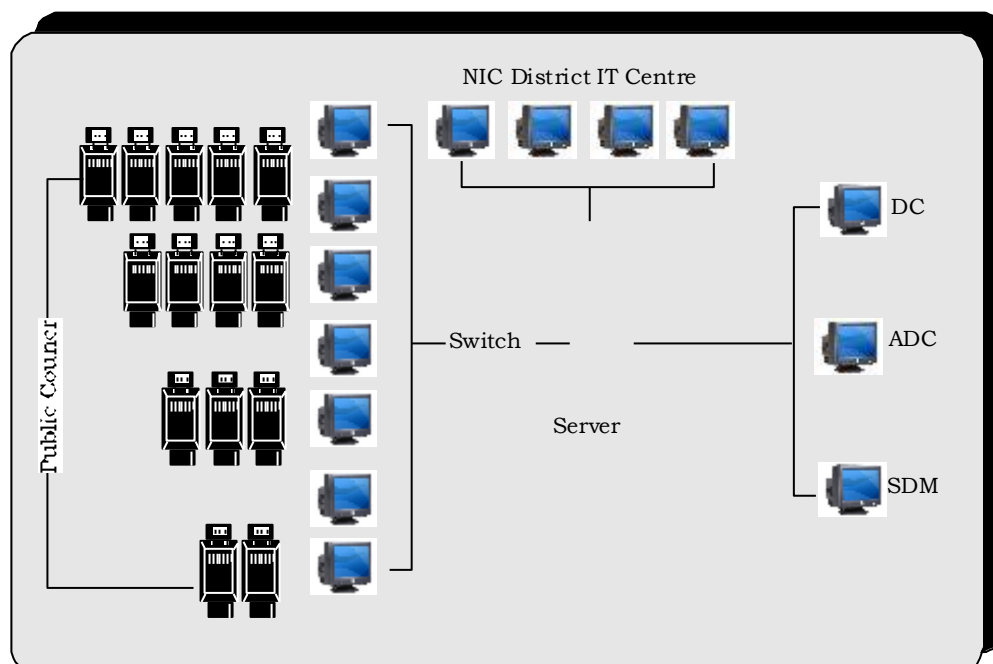


Figure: Technology Behind Ekal Seva Kendra

Operational Aspects

1. Sarathi

A software, which provides comprehensive and integrated solutions for issue of driving licence has been implemented. Driving test data is blocked on the applicant's choice and the modularity of software has been designed to ensure the delivery of the licence in stipulated time period and to produce cash/audit statements.

2. Vahan

Vahan Software has been implemented for the solution of vehicle registration relation works. Software has provision of online security of the documents and booking of vehicle inspection data on the applicants choice. Several types of MIS reports for the District administration and other agencies like police, insurance companies, etc. can be generated instantly.

3. Certificate

Software for issuance of certificates of Caste, Birth and Death, Haryana Residence has been implemented. The Kendra is issuing certificates instantly on submission of verified document and in case of non-verified documents, certificates are being issued with stipulated time period after getting the Verification concerned officer.

4. Nakal

Nakal of Jamabandies, Mutations and Khasra Girdawari are being issued from this counter within stipulated time frame. Nakal are being generated from the online computerized data available as well as from the manual records.

5. Passport

Computerized file reference number is being issued to the passport applicants and after scrutiny these are forwarded electronically to Regional Passport Office for issuance of passport.

Self Sustainability

The entire project has been set up without any budgetary allocation from the Government. It is funded by District Red Cross Society on the lines of 'Haryana Registration Information System (HARIS)'. All running expenses including those of manpower are also being met by the District Red Cross Society. For this purpose, a nominal contribution from user is being charged for each service.

These service charges are inclusive of facilities such as on the spot blood group test wherever applicable, government prescribed forms, photographs, Lamination of documents, booklet of traffic rules and symbols, file folder etc.

Sachivalaya Vahini or E-Governance in Secretariat

Introduction

Secretariat is the highest policy making body and the apex seat of the State government. It is a repository of a wealth of governmental knowledge, which needs to be managed efficiently for achieving the desired objectives. In its capacity as the apex policy making and decision making authority at the State government level, the Secretariat has to deal with a large number of departments, organizations and individuals. This process involves enormous amount of communication, besides holding, maintaining and processing of large volumes of data files or other formats. Conventionally, all the files were created and maintained manually resulting in very slow and inefficient process.

Secretariat LAN (Local Area Network), the fibre optic Intranet set up by the Directorate of Information Technology (DIT),* Government of Karnataka, Bangalore, connects more than 1000 computers spread across 40 Secretariat departments catering to the needs of 6000 secretariat staff. To promote e-governance by facilitating the policy and decision making process and to introduce the concept of inflammation and Knowledge Management making use of this large network, national Information's Centre (NIC), Karnataka State Unit, Bangalore, designed, developed and implemented a suite of software packages in all the 40 departments of the secretariat, under the title Sachivalaya Vahini,

Services

Patra-The Letter Monitoring System (LMS)

This is a software package for managing the large number of letters received in any department of the secretariat. The letters can be scanned and moved from desk to desk and even across departments for action to be taken, till the letters are filed or disposed. Records of dispatched letters can also be maintained.

Highlights

- Status of letter received (where it is and why it is pending) can be known without any delay.
- The entire movements of the flow of the letter can be traced, including action taken without any delay.
- The file note in which the letter is filed can be identified.
- Any query based on any parameter of the subject of the letter can be answered.
- Pendency of all letter received can be ascertained, department wise, section wise, or officer wise.

Kadatha-File Monitoring System (FMS)

This is a decision support system aiming at monitoring, tracking and helping in the speedy disposal of files, thereby increasing the efficiency of the workforce. Electronic files can be moved from desk to desk and even across the departments.

Highlights

- Nothings on the file can be entered.
- Draft documents can be attached to the file.
- Electronic file can be moved from desk to desk and across departments.
- Status of file, for instance, where the file is pending and for what reason it is pending can be known instantaneously.
- All the letters in the file can be viewed.
- Individual office wise, sections wise, departments wise and across the departments pendency of files can be obtained.
- All the registers and records that are maintained manually can be generated without any effort.
- Pendency of all letters received can be ascertained department wise, section wise or office wise.
- Extensive queries based on any parameter of the file or on actions taken on the file can be answered.

Mokaddame-Court Case monitoring System (CCMS)

This package is to monitor the court cases in which government is the respondent of petitioner. Cases received, Petitioner/respondent details, court orders, cases put up for hearing on a particular day and the line, can be managed efficiently with this system.

Highlights

- Provides information about all cases filed against the government and by the government.
- Cases due for hearing on a particular date can be known immediately.
- Status of any case can be known without any time delay.
- Orders issued by various courts on a particular subject are made available.
- Various reports-courtwise, subjectwise, respondentwise, petitionerwise, casewise, etc, can be generated.
- Automatic generation of notices with regard to appointment of government advocates, and instruction to prepare draft Statement of objections.

Aayayaya-Budget Monitoring System (BMS)

This module aims to arrive at the budget estimate and to monitor the proposals, once the budgeted amount is allocated to the department.

Highlights

- Appendix-B and all the required annexure which assist in the salary related budget estimation process can be generated.
- head of account wise budget annual allocation is maintained.
- Sending/updating of proposals and issue of government orders can be carried out.
- All the proposals pending for approval of Finance Department can be viewed.
- All the proposals pending for approval of Finance Department can be viewed.
- Departments can analyze supplementary budget requirements, with drawal/surrender of proposals and reallocation of funds.
- Monthly Multi-level Review (MMR) of Expenditure can be processed.

Sibbandi-Personnel Information System (PIS)

Personnel Information System capture all the details of the employees as recorded in the service register.

Highlights

- All details of the employees can be searched on under the name of the employee.
- Entire service history of employees can be viewed in a very short time.
- Employees leave details can be monitored.

- Employees who are going to retire for any given period can be assessed.
- Employees who have been promoted/not promoted for any given period can be found out.
- Repository of Employee information can be generated.

Customer Support System (CSS)

Through the Customer Support System, any complaints regarding Hardware, Network, Application Software System Software, etc can be lodged, and the system will help monitor the complaints by giving online information.

Highlights

- All complaints regarding Hardware, Network, Application software, System Software, etc. can be lodged.
- When the concerned person who has to rectify the problem logs in, the complaints related will be displayed.
- Once the problem is attended or rectified, the status can be updated.
- The call will be cleared, when the customer who has lodged the complaint authorizes his satisfaction regarding the rectification of the problem.
- Numerous reports like calls registered for a given period, calls which have not been attended, calls which were attended, rectified, etc can be generated, which will help in the monitoring process.

Special Features

Integration

All the above discussed applications are integrated with one another so that one application can access the data of another application. Depending on the access rights of the user who logged in, different menus will appear.

Dash board

This will give an overall view of pendency and status of letters, files, court cases, budget proposals and personnel related details.

Single login

Once a user is logged into one application, he can access any other application without logging again. From any application, any other application can be called.

Management and Security Features

- The applications run on Windows 2000 Advanced Server/SQL Server Enterprise Edition/VB 6.0 MS Access 2000 platform
- Centralized database server is maintained to facilitate easy maintenance and backup.

- Active directory services are used for deploying applications to all clients automatically.
- Automatic DSN creation through application.
- Database accessibility is through database users with restricted permissions.
- Database updation is possible only through 'Application Roles'.
- Application accessibility is through domain login.
- Active directory group policy is used for access control.

Training

NIC has trained more than 6000 secretariat staff of all levels on the application packages. Over 80 demonstration of the software were given to various departments, and based on the feedback received, software features were enhanced.

Implementation Status

- Patra. Letter monitoring System:** This is being used by all departments. At present there are more than 3,00,000 letters being monitored through the system.
- Kadatha. File monitoring system:** This is being used by all departments. More than 2,00,000 files movement and pendency are being monitored through the system.
- Mokaddame. Court Case monitoring system:** Its usage was started by forest, Cooperation and Finance Departments.
- Sibbandi. Personnel Information System:** All service register details of around 3,000 secretariat staff have been entered and data validation is going on.
- Customer support system:** This was launched to register complaints regarding secretariat LAN related hardware and software problems and to initiate remedial action.

Future Enhancements

- Web enabling: All queries related to the above package of general public interest are being web enabled.
- Knowledge management. A knowledge base consisting of all Acts, Rules and Regulations, Circulars, Government Orders, etc. will be set up so that any application can access this.
- Knowledge base to assist in the decision making process.
- Public grievances system: A general public grievance system is being prepared.
- Customer feedback system. A feedback mechanism for the customer (or user) of the software is proposed to be provided.
- Stores management system. A management information system for stores is also proposed to be developed.

- (vii) Organizations structure. Organizations structure of every department will be made available along with complete information of the departments with link to employees information, department schemes, and any other relevant information.
- (ix) Notice board. Any Circular or Order issued by any official or department will automatically be made available to the concerned persons.

Conclusion

In this case study we studied the details of suits of software applications in Karnataka State Secretariat at Bangalore, such as Letter Monitoring, File Monitoring, Court Cases Monitoring, Personnel Information System, Customer Support System, including training implementation details. They system can also be adopted in other state secretariats.

Bhoomi

Introduction

Bhoomi is a project jointly funded by the Government of India and the Government of Karnataka to digitize the paper land records and create a software mechanism to control changes to the land registry in Karnataka. The project was designed to eliminate the long-standing problem of inefficiency and corruption in the maintenance of land records at dispersed and poorly supervised and audited block-level offices known as "taluka" offices in South India and "tehsildar" offices in North India. The project development and implementation was done by National Informatics Centre. Many experiments with computerization have failed due to corruption and other factors.

Implementation of Land record computerization has been difficult in India. Bhoomi succeeded because there was a champion who worked a 15-hour day for over 12 months, devoting 80% of his time to the project. Minimizing resistance from staff by harnessing political support was an important contributory factor. Extensive training coupled with a participatory style also helped to diminish resistance."

In Karnataka State, the land records were earlier maintained through a manual system, involving 9,000 village accountants, each serving a cluster of 3–4 villages. Nearly 2,500 bank branches in Karnataka loan approximately Rs. 40 billion to farmers as working capital every year. The concept of Computerized Land Records Management Process was first introduced in Karnataka through the BHOOMI initiative in 2001 to bring in overall transparency, effectiveness and ease in the management and maintenance of the Land Records through automation of various processes.

The major objectives to be fulfilled by the Bhoomi project were:

- Facilitating easy maintenance and prompt updating of land records
- Making land records tamper-proof
- Allowing farmers easy access to their records
- Collating the information to construct a data base regarding land revenue, cropping pattern, land use, etc
- Utilising the data for planning and for formulating development programmes.

Through this initiative, Revenue Department in Karnataka has computerized 200 lakh records of land ownership of 67 lakh farmers in the state.

In 2010, concept of electronic integration of BHOOMI and KAVERI (registration Software) was conceptualized and piloted in five taluks namely, Bangarpet, KGF, Malur, Kolar and Tumkur. Overwhelming success in pilot resulted in rolling out of BHOOMI-KAVERI integration all across Karnataka covering 244 out of 247 Sub Registers Offices spanning over 200 taluks except three taluks due to reasons such as non availability of connectivity, infrastructure etc.

Land acquisition is another important activity which results in updations to RORs at various stages of acquisition process. BHOOMI was electronically integrated with BHOOSWADEENA software in 2011. Project is running successfully in all 52 sub division offices of revenue department and 27 Special Land Acquisition Offices across Karnataka.

Electronic integration of BHOOMI with Banks was started in 2012. Approximately 1400 branches of 29 banks across Karnataka are using the facility to raise request for creating or removing charge and also enhancing the liability on agricultural lands belonging to farmers who are availing farm credit.

Land Record

In Karnataka, there are 17 million land record documents covering 20 million farmers. Karnataka has 27 districts with 177 taluks covering around 800 hoblis and 29,000 villages. This land record document has valuable information, which includes ownership tendency, cultivation, irrigation, tree, liabilities, crop, and soil details including the details of Government lands.

We need land records for the following reasons:

- Office Land Records is required to be produced for obtaining crop loans from any recognized financial institution as a proof of ownership and existing liabilities.
- Traditionally, people are satisfied with their ownership on land by having the Land Record issued by the Revenue Department.
- Most of the legal disputes are settled based on the time sequence of the Land Records issued for a survey number.
- The government has a duty and responsibility to protect the ownership of land based on its information maintained in 'Land Records Registers'.

Drawback of manual system over computerized system

The manual system also had the following drawbacks:

- The issue of Land Records depends on the availability, mood and interest of the V.A/Pattwari.
- Large-scale corruption
- The manually written Land Records may not be legible.
- Land Records Registers are not maintained properly.

- It takes a long time for reproducing Land Records for the succeeding year after incorporating the current year crop details, liabilities, changes in ownership or cultivators which happen through mutations.
- Very difficult to prepare cross tabulated registers or reports.

Now, in the computerized system, the farmer can collect his land records within a couple of minutes at the Land Records Kiosk, without any difficulty or delay.

BHOOMI-land records management software

BHOOMI, the software for land records management, has been fully designed and developed in-house by National Information's Centre, Bangalore, Department of Information Technology, Ministry of Communication and Information Technology, Government of India, using state-of-the-art technology. One year effort has gone in bring out the first version of software modules. The following features and incorporate:

- Bhoomi is an online system to carry out mutation on the live data,
- It has built-in workflow automatic, which moves transactions from one officer to another one the system, as shown in Figure C9.1.
- The process of mutation of the BHOOMI is fully synchronized.
- It also facilitates scanning of the field mutation order passed by revenue authorities and notice served on the public and storing into database so that it can be refereed easily in future for various purposes.
- It has also been integrated with Fingerprint (Biometrics) technology to ensure better authentication instead of traditional password system. This enforces the concept of non-reputation.
- The software is in local language (Kannada) for easy use by officials.
- Various analysis report can be generated in text format and also viewed in graphical style.
- Approval of the transactions is on first-in-first-out basis.
- Transactions will be pushed into next approval authority if it had not been approved by the concerned personnel after specified period.

Modules of BHOOMI

It has two modules for public interface:

- One module is used by revenue officials of Land Records Centre so issue the land records documents on demand from the public and accept the request application for mutation from the public.

- The other module runs on a Touch Screen Kiosk, set up at taluka/block office. This module is easily operable by even the personal farmer having little knowledge of computer as this is developed using "Keep it Simple" (KIS) concept.

Implementation process of BHOOMI

The implementations process of computerization of land records starts with digitizing the legacy data. This is a laborious process as there are around 1 lakh documents in each taluka/block, whose data is to be fed correctly and properly into the computer. The District Administration has assigned this work to private data entry agencies. The Village Accountants go to the premises of data entry agency and ensure that data is fed correctly from the good old land record books. This is an iterative process. The checklist of documents will be printed by the data entry agency and submitted to the Village Revenue Official. The village Accountant verifies the printed checklist with that of original record and corrects the checklist accordingly. The correction of wrongly entered data will be carried out by the data entry agency. Once the data is corrected, the agency will take the final print of land record documents. Again the Village Revenue Official verifies the printed data for correctness. The process repeats till all the data entry errors are eliminated. The data entry agency hands over the digitized data in CD form to the Revenue authority. The Revenue official cross-verifies the final printed land records documents with the manual documents on random basis.

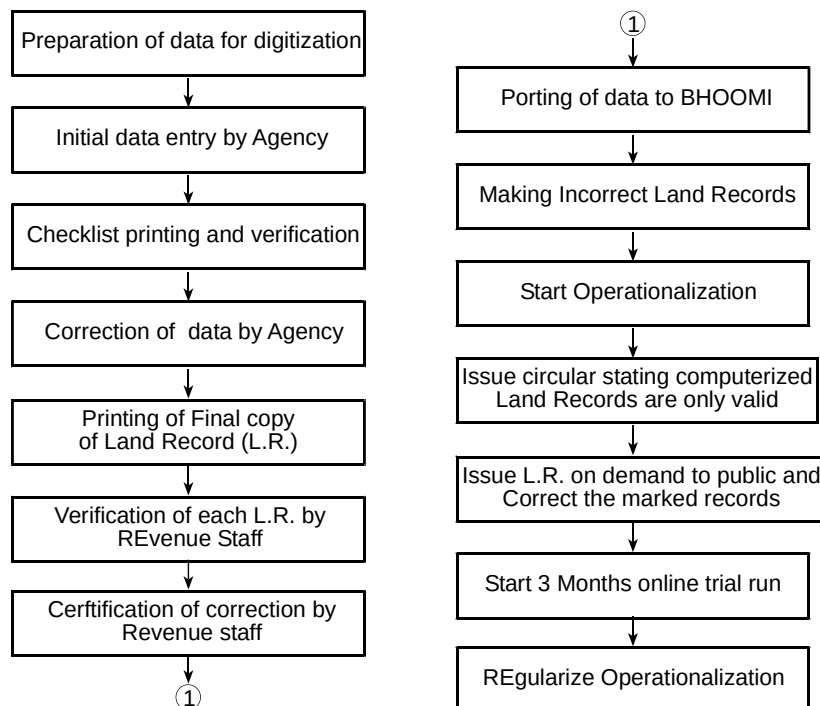


Figure: Implementation process of BHOOMI

Conclusion

In this case study, the Bhoomi project has been surveyed in detail with all aspects of introduction, objective, design and implementation of the software. . BHOOMI is a great success. On the lines similar

to Bhoomi, many other States such as Tamil Nadu, Andhra Pradesh, Haryana, Gujarat, and Madhya Pradesh have also successfully implemented Land Records MIS.

IT in Judiciary

The judiciary is a system of courts which interpret and apply the law. The role of the courts is to decide cases by determining the relevant facts and the relevant law, and applying the relevant facts to the relevant law. The **Indian Judiciary** administers a *common law system* in which customs, securities and legislation, all codify the law of the land.

Technology has proven its role in enhancing efficiency, improving access, as well as encouraging transparency, accountability, and adherence to timelines in dealing with legal matters.

India has taken a variety of measures to automate processes within the judiciary with an aim to improve efficiency and eliminate the time lost in unproductive work. Some of the initiatives are: easy accessibility to online case records and previous judicial pronouncements, reducing dependency on stenographers by utilising voice recognition software, usage of video conferencing facilities to increase the reach of courts, case management systems, e-registry of court, automated preparation of cause-lists, phasing out physical records through e-filing and e-submission of documents, etc.

Judicial applications of IT

In 1990 a project named COURTS (Court Information System) was launched for the benefit of the entire legal community. This project was commissioned for streamlining registries of community. This project was commissioned for streamline registries of various courts, Subsequently, all High Courts have been computerized and Web enabled both locally and nationally through NICNET.

The following applications have been successfully implemented at the Supreme Court and also 18 High Courts in the country. The end result of these applications has been a lot of convenience, improvement and speed of legal service for the litigants who are common people of India.

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Case Status (<http://www.caswstatus.nic.in>)

This web site provides Supreme Court's pending and disposed case status information to litigants/advocates on the Internet. Case-status gives the latest information with respect to the status of a case which could stand as Disposed, lower, court details, party and advocate names, date on which last listed, waiting position, subject category along with exact verbatim of the text of the Court's order. Pending case status can be accessed through Case number, title, advocate names and lower court details.

As soon as a case is filed in the court, its case-status is available on the web, giving filing details such as dairy number, date of filing and the like. Litigants can find out whether their cases have been filed in the court of not without contacting the advocates. Case-status also provides all orders pertaining to a case delivered so far by the court. The litigants can maintain their own case files by downloading all order pertaining to a case. Case-status receives hundred of hits every day.

An advocate can download all his cases pending and disposed and maintain his own database.

Impact

Response to the 'Case-status' from the public is overwhelming, as pending cases information is available at his/her residence. It avoids the litigants to come all over to Delhi from their place to find out the status of their pending cases. The litigants need not find the status of their pending case cases on phone as is the usual practice. Very authentic information, which contains the latest Order of the court is available to the litigants at their place.

JUDIS (<http://www.judis.nic.in>)

This is the Judgment Information System (JUDIS) on CR-ROM consisting of complete text of all reported judgements of Supreme Court of India from 1950 to 2000. The judgements of 2001 onwards are available on the Internet. JUDIS CD is available on a membership basis. It is marketed by NICS (NIC Services Incorporated), New Delhi.

Judgments can be accessed through party name, advocate, date of judgements, and more importantly, through free text search. Free, text based search enables thusers to retrieve all relevant judgments of a particular subject. Judgments are available on the web site within 24 hours of their delivery in the court.

The judgments of the High Court of Delhi since 1999, the High Court of AP since 1999, the High Court of J and K since 2001, and the High Court of Orissa since 1985 are available on the web.

Impact

- An ordinary litigant can find out precedents of a case on his/her own.
- Saves lot of time of judges and advocates in locating precedents.
- Savages lot of space in maintenance of a law library.

It proved to be very useful advocates who cannot afford to have his/her own Law library. As JUDIS CD contains complete text of the judgments, then is no need to refer to any law journal for findings precedents.

Cause lists on Internet (<http://caselists.nic.in>)

Causelists are schedules of cases to be heard by the courts the following day. The Causelists of Supreme Court and all High Courts are available on NIC web servers in the web site <http://caselists.nic.in>.

Immediately after generation of the causelists, most of the courts cyclostyle the stencils cut from the printers attached to the servers for generating thousands of copies running into a few tasks of pages every day Due to this, the courts take a lot of time for generation and supply of the causelists to the advocates at their offices or residence. Usually, the advantages receive the cyclostyled copied of a day's causelist not before 8 p.m. Some High Courts copy the caselists data on a floppy and print thousands of copies. This process costs each High courts lakhs of rupees every every. By making the Causelista available on the Internet, on High Court is incurring any expenditure as they are using the already available infrastructure and the software of NIC.

Features

- Advocates can generate their own causelist consisting of his/her own cases.
- Retrieval can be made through the name of either petitioner or respondent.
- Courtwise list can be generated.
- Judgewise list can be prepared.
- Entire causelist can be printed, if required.
- Case number wise access is possible

Impact

- Advocates are able to receive the cauelists almost immediately after the court hours
- Advocates can generate their own cauelists which will contain only their cases, thus making them avoid to go through hundreds of pages to locate their cases.
- As the application is available on the Internet, the litigant public can easily find out whether their cases are coming for hearing or not, without bothering the advantage.
- Some courts are considering to the to generation of copies of cauelists, as mot of the advocates are dependent on the internet version of cauelists. Thus the courts can save goods amount of money on annual basis.

Daily Orders on Internet (<http://www.dailyorders.nic.in>)

The daily orders of Supreme Court and the Delhi Higher Court are available on the Internet. As soon as the orders are signed by the judges, they are made available on the Internet. These order are only for information of the litigants, Official copy of the signed order has to be obtained only through normal channels. They daily orders of other High Courts re constantly updated. It is an instant hit with the litigants and advocates. The database are maintained by the respective Courts.

This is the easiest way for litigants to get a copy of the latest order delivered in the courts, from their residence/offices. The free text based search enable users to access relevant order of the court on the same subject,. It also helps the users in accessing order without knowing the case number, or party name.

Impact

- Litigants can have access to the signed orders from their own place.
- Enhance confidence in the judicial process.
- Saves lot of time and expenses on travelling to the court to obtain a copy of the order.

Applications Implemented in the Supreme Court of India

Following are some of most important applications implemented in the Supreme Court, which helped the Registry in streamlining its routing activities.

Judicial Wing

Filing Courter computerization

Fresh cases are filed only before the computerized filing courters. As the advocates stand in queue for filing cases before the counters, the data entry operator enter preliminary details of the case such as party names, advocate details, etc. required for registration. Filing receipts are generated, court fee and time limitation are calculated automatically and automatic registration takes place, Filing information is immediately available on the Internet.

List of Business Information System (LOBIS)

This contains pending and disposed cases since 1990. The size of the database is about 5 lakh records. It is near time-critical application as the causelists are to be generated by the computer system by the close of the court's closing hours every day. This application also uses bunching/grouping technique which enabled the registry in bringing down the pendency in the Apex Court.

Court's Order/proceedings Computerization

This is about computerization of day-to-day orders delivered in the court. The software enables the Court Masters in speeding up the process of typing orders/judgments by providing the preamble, which includes the item no. of the case in the causelist, name of parties, advocate names, High court/lower court details, name of judges before whom cases are listed, date of order and so on, on the screen. The Court Masters need to type only the order part note by them in the courtroom.

Courtnic: Judgement Information System (JUDIS)

Record room computerization

This enables track of keeping files and printed records consignment, weeding of files, printed records and their maintenance, movement of files, and tracing of files.

Statistical reports

Every month it generates 22 pages of statistical reports on institution, and disposal and pendency of variance types of cases in the Supreme Court Of India.

Web site of Supreme our of India (<http://ss\remecourtfindia.nic.in/>)

Causelist on Interest

All the requists inputs about Supreme Court are available in its web site. The causelist is available on the Interest for public use.

Judicial Sections

All judicial sections are provided with computer systems for updating the case databases pertaining to their sections and minimizing movement of files, apart from generation of notices and dismissal letters Registration of cases, Disposal of cases, and finding status of cases.

Library information system

SUPLIS-Case indexing. In contains citation information of all reported cases decided by the Supreme Court from 1950 onwards. Cases can be retrieved through case number, party name and subject category. This is available on the Supreme Court's web site.

Current contents. This keeps trace of all legal and judicial articles published in journals subscribed by the library of Supreme Court. This is also available on the Supreme Court's web site cataloguing. This helps maintain details of all books available in the judges' library of Supreme court and supports 'Issue and Return'.

Administrative Wing

This includes

- Pay roll, inventory control system, personal information system and monitoring or recovery of loans.
- Monitoring of bank guarantor's cases, monitoring of payment to petitioner/ respondent for court ordered cases.

High Courts computerization

NIC took up computerizations of all 18 High Court of India and 10 benches of the High Courts on the lines of the Supreme Court's computerization. It implemented the List of Business information (LOBIS) in all High Courts. All High Courts' Causelists are also available on the Internet. Most of the High Courts have opened query counters along with filing counters for providing case status information to the litigants and advocates.

Facilities

- All High Courts are provided with computer hardware with back-up facility.
- All court rooms are equipped with terminals and printers.
- All court rooms are connected to LAN.
- Some High Court Judgments and orders are available on Internet.
- Most High Courts have their web sites.
- Causelists are generated automatically.
- Posting of cases to various courts are done by the system.
- Bunching/grouping is done.
- Court fee and time limitations are calculated automatically.
- Notices are generated.
- Computer based filing counters are opened.
- Filing receipts are generated for reference for advocates.
- Query counters are available.
- JUDIS and COUNTING are available.
- JUDIS and COURTNIC are available.

- All High Courts are connected on NICNET/Internet.
- Day-to-day judgements and order are stored, on computers.
- Most of the High Courts' libraries are computerized.
- Administrative and account applications have been computerized.
- Some of the High Courts have FTDMA/IPA V-sat based Internet connectivity.

Impact

- Registries' day-to-day work is streamlined.
- Filing process is streamlined.
- Litigants can get cause-status information at query counters.
- Causelists are generated on time and accessible on web.
- Court orders are generated with case.
- Court orders are generated with case.
- Notice are generated without delay.
- New work culture got introduce

District Courts' Computerization

In 1997, NIC took up the computerization of all 430 district courts in the country on the lines of the High Courts Computerization Project. The basic objective of the project are to:

- Provide transparency of information to the litigants and advocates
- Help the judicial administration streamlining its activates
- Provide judicial and legal database to the District judges.

NIC provides three-level training programmes to the district court official. The three levels are: World processing, training on application software package for district courts, and training to district court judges.

All officials have been training on the District Court information SYSTEM developed for the computerization of district courts. This software takes care of all aspects of district court needs.

Subsequently from 2001 pilot district courts in states such as Andhra Pradesh have been provide with a LAN (Local Area Network) in the entire court with 50 nodes. Several online applications such as Cause list generation office automation, A diary Personnel Information System. Payroll,etc (on the lines of high court) have been implemented.

Impact

Some of the district courts are now able to:

- General causelist.

- Store judgements on computers.
- Generate notices automatically.
- Generate certified copies
- Monitor case flow.
- Assist in caveat matching Post Cases in a transparent manner
- Access Supreme Court judgements.
- Account e-mail and internet.
- Give Case-status information query counters to litigants

Conclusion

In this case study, we have presented a details view of the efforts of NIC in computerization of the Supreme Court of India, Various High Courts and also the district courts. Various judicial web sites at various levels of Judiciary are also referred. This is a major work successfully implemented and is like to be emulated by Judiciary in other developing courtiers also.

E-Khazana for Government Treasury, Andhra Pradesh

Introduction

In tune with the policy initiative taken up by the Government of Andhra Pradesh for effective utilization of Information Technology as part of e-Governance project, National Informatics Centre, Hyderabad has taken up online computerization for the its Districts Treasury (DTO), Sub Treasury (STO) and Pay and Accounts Offices (PAO) Offices in 330 locations across the state.

E-Khazana is an online application that takes care of entry level validation and budget control and pre-audit rules at Auditor level and finally for issue of cheque/pass order at passing level. After reconciliation with bank, monthly accounts will be generated for submission to Accountant General.

The Administrators' role is very important in this application as be will be responsible for

- Assigning the roles.
- Watching budget expenditure.
- Head of accounts, and
- Drawing and disbursing officer.

It is an intranet applications that works on linux/Windows platform with backend as postages SQL 7.3 and front-end is PHP 4.1 (Hypertext Pre Processor).

The Treasury System in AP

Director of Treasury and Accounts (DTA) is the administrative head of treasuries and accounts department.

STO office is responsible for compiling all the payments through DDOs and receipts through challans. This data is sent to DTO on a daily basis. They also send Pensioner details and Class IV GPF details to DTO on a monthly basis. There are around 300 STO offices at the sub-district level in the State.

DTOs in turn compile all the accounts (Payments and Receipts) processed through its associated Sub Treasury Office (STO) and are located at the District headquarters. They are also responsible for sending reports to AG office and DTA. DTO offices in the state are located at 23 District Headquarters.

Office of the Directorate of Treasury and Accounts (DTA) is responsible for compilation of all Payments and Receipts processed through DTO (District Treasury Office) on daily basis and sending reports to Finance dept. (Budget section), CM's office, and to other Heads of Department (HoDs).

Figure 5.2.11.1 shows the hierarchy of offices in the Treasuries Department, in the Government of AP.

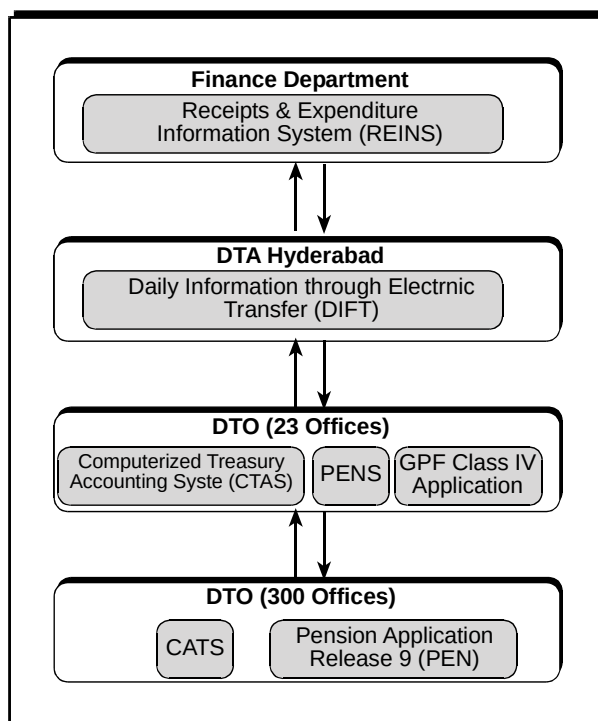


Figure: Hierarchy of offices in Treasuries Department

Introduction to Functions of Treasury

Functions of the Treasury are broadly classified as:

- Treasury functions
- Accounts functions

Treasury functions

These functions can be summarized as follow:

- Receipts and payments made either by cash (bills or cheques) adjustments (bills or cheques)

- Conducting Government's financial transactions such as payment of salaries, other payments like TA, LTC, contingent rents, loans and advance, receipts of Government revenue, etc.
- Conducting or cash transactions through non-banking treasuries and banks.
- Payment of pensions
- Pre-audit of bills and post-audit of vouchers
- Reconciliation of the valuable articles such as election boxes, examinations papers, department cash chest treasuries trove, and the like.
- Receipt, safe custody and sale of various kinds of stamps.
- Payments of Government Securities such as Stock certificates, bearer bonds, promissory notes and income tax deductions
- Exercise treasury control over all departmental financial transactions with reference to Budget.
- Maintenance of GDF account of last grade employees
- Maintenance of Employees Welfare Fund Account of employees
- Regulation of appointment to the public service through Act-2 of 1994.
- Rendering of daily figures of Finance department
- Rendering of classified accounts to the AG
- Maintenance of deposit accounts including GO No. 43.
- Maintenance of User charges Accounts.
- Submission of PMES information to Government
- Enforcing accountability on DDOs and reporting to the Government as per GO NO. 507.
- Maintenance of employee census database.

Accounting functions

Accounting functions include the following:

- Rendering financial advice to head of the department
- Preparation of budget estimates of the respective departments.
- Maintenance of departmental accounts in the respective departments
- Reconciliations of the receipts and expenditure of the department
- Processing of the pension cases of the respective departments

- Internal audit of the Department Unit Offices and follow-up action for settlement of the audit objections.

Reports generated by treasuries and accounts department

- Classified Accounts to AG-STO/DTO level
- GO No. 507–STO/DTO/RJD/DTA/PPO/APPO
- PMES-STODTO/RJD/DTA/PPO/APPO
- Daily figures-STO/DTO/DTA
- Receipts and Payment Reconciliation-DDO-HOA wise
- Pensions/strong room stamps
- Budget vs. Expenditure –DDO wise and object headwise
- Non-Banking Sub treasuries
- Class IV GPF/EPF
- Deposits Accounts and GO No. 43
- User Charges
- Employee Census
- Any other report required with available data.

System Overview

Budget authorization

The DTA, after exercising check on Distribution Statements furnished by HoD's with reference to Budget Releasing Orders (BRO) and Budget Estimates (BE) approved by the Assembly, issues authorizations to the DTO's concerned for Budget distribution made by the HoD.

Redistribution/Reauthorization

At any District, the subordinate controlling officer or the Head of the District, further distributes the Budget down the line to each Drawing officer and sends a copy of this budget distribution statement to the DTO. The DTO, after exercising proper cheques with reference to the Budget authorized by the DTA, issue, re-authorization to the STO concerned.

The Treasury Officer permits the Drawing and Disbursing Officer (DDO) to draw the money to the extent of authorized Budget provision.

Honouring of Claims against government

The DDO from each offices office presents their claims to the Treasury department for withdrawn of money for various purpose like "Payment of salaries, implementation of various schemes, meeting of day-to-day expenditures in the officers, etc." Based on the nature of the claim, different bill forms are used.

Treasuries honour these claims after scrutinizing the admissibility of the claim, financial powers of the drawing officer, sanction orders required, the genuineness of the claim, etc. besides availability of budget. After thorough examination of the claim presented by the DDOs, the Bill is passed or the Bill is returned when it is not in order duly quoting the authority for its return.

Acceptance of receipts

Receipts that accrue to the Exchequer are remitted through an instrument called challan. This challan form is used for all sorts of receipts like tax revenue or fees to be paid or a recovery.

Treasury checks the challan whether HoA is reflected on the challan with reference to the purpose of remittance and the Department on whose behalf the remittance is made. An entry is made in a register and serial number of the entry is noted down on the challan with date. The Treasury officer in token of having checked the same appends stamped signature on challan and then only the challan is accepted in the Bank for remittance.

Verification and certification of bank scrolls

Bank scrolls received together with the paid vouchers, cheques, or challans, are verified on a daily and monthly basis, RBDs are arrived at and certified before finalization of the daily/monthly accounts by the treasuries.

Rendering of daily and monthly accounts

For any receipts and payments for the day, every bank branch sends both challan's and the paid voucher's list to the concerned treasury. Each day, STO sends statements of receipts and payments to the DTO, who takes into accounts the receipts and payments in the District and Accounts rendered by its sub-offices. DTO prepares daily accounts and submits to the DTA. DTA in turn furnishes the consolidated daily figures to the Government.

All sub-treasuries, at the closure of the month, daily accounts for the months are classified into sub-accounts as required by the AG's and the same are furnished to the DTO. Treasury at District consolidates them as Sub-Accounts and Main Account and Furnishes to the AG on or before the specified due date.

HoA-wise figure are submitted to the DTA, who in turn submits consolidated figures to the Finance Department Consolidation and rendering of daily/monthly accounts are handled using software package called C-TAS (Computerized Treasury Accounting System) and DIET (Daily Information through Electronic Transfer) which were developed in-house.

Supply of Stamps to Sub-registrars in Districts

Treasuries undertake another important revenue earning function of the State Government. The Treasury, as a Local Deposit, stores the Non-Judicial and Judicial Stamps required for registration and other purposes.

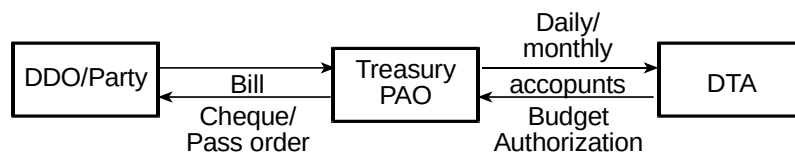
Treasuries keep sufficient stocks of stamps by placing by placing timely indents to the Inspector General of Registration and Stamps, and supply the same to the Stamp Vendors and to the Sub-Registrars. They render Plus/Minus Accounts every months to the AG and to the Inspector General of Registration and Stamps. Treasury Officer conducts annual stock inspection of stamps held by Sub-Treasuries and the Sub-Registrars.

Preparation of budget estimates

The Department prepares Budget Estimates, by collecting Number statement and other estimates from all the District Units. It submits the same to the Finance Department duly taking into accounts the Budget estimates and revised estimates of the previous year, the expenditure estimated under various heads for the current financial year strictly adhering to the norms prescribed.

Context Diagrams

The context diagram for Treasuries is depicted here as Figure C11.2 to show the system level context diagrams.



Overview of District Treasury Office Functions

Figure presents an overview of the DTO functions:

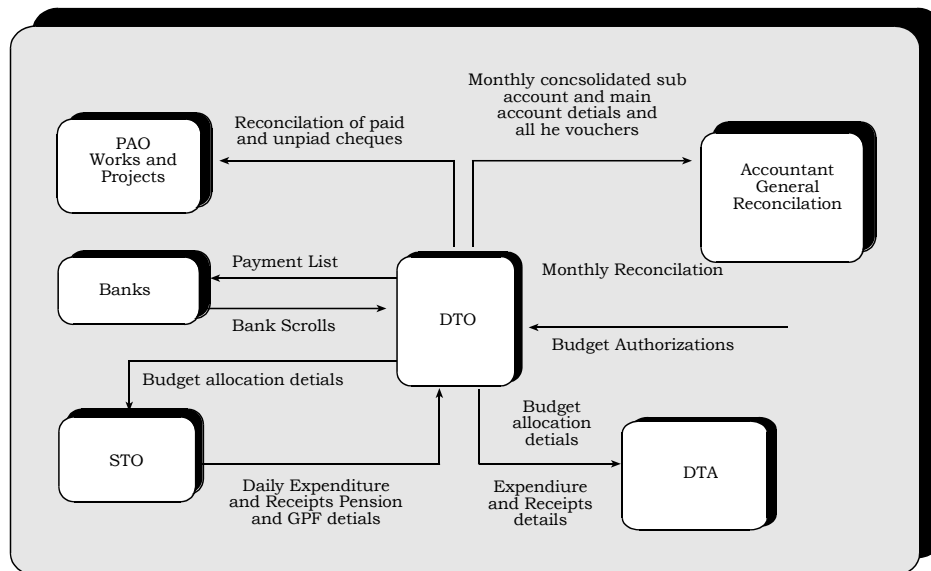


Figure: District Treasury Office Functions

Core Functions of POA–Twin Cities

Let us summarize the core functions of payment and Allowance Office (PAO) in the twin cities of Hyderabad and Secundarabad:

- Payment of regular pay and allowance
- Regulation of pay and allowance of AIS officers.

- Maintenance of GIS of AIS officers.
- Regulation of Leave Salary and pension contribution for AIS officers who are on foreign service.
- Maintenance of Class-IV GPF accounts.
- Reconciliation of paid cheques.
- Preparation of Civil list of AIS officers during January and July.
- Payment of salaries through Electronic clearing system to the Honourable ministers, Hon'ble High Court Judges, Senior officers in secretariat and Gazetted Officers, Heads of Departments and all Government employees in the Twin Cities.
- Rendering of Accounts to Accountant General/Finance Department Pre-audit of claims
- Monitoring the submission of the DC Bills for the AC Bills drawn
- Submission of PMES Information to Government
- Enforcing accountability on DDOs and reporting to the Government as Per GO NO. 507.
- Maintenance of employee census database.

Budget Allocation from Finance Department

Subordinates of each department (DDOs) send budget proposal regarding their needs to their respective Head of the Department. HoDs of each department verify and send it to their concerned department in Secretariat which in turn submit the proposal to the Finance Department.

On receiving the proposals, the Finance Department allocates the Budget and sends the Budget Allotment information to the HoDs of each department. On receiving the Budget allotment information, HoDs of each department prepare Redistribution of Budget Information for their subordinates and will send one copy of Redistribution of Budget Information to their Subordinates (DDOs) and three copies to DTA. ADA keeps one of the three copies with itself. DTA sends the second copy to the Audit Officer (DAO for districts and PAO for Twin Cities) DTA sends the third copy (Authorized and Stamped) to the Subordinates (DDOs).

Scrutiny of Bills and Approval by PAO

The DDO of the concerned department submits the bill of government expenditure at the PAO and receives a token as an acknowledgement. The validity of the bill is checked as per the government rules for the specimen signature and arithmetic accuracy, in the three states by Auditor, Superintendent, APAO/Deputy PAO at PAO. If the bill satisfies the stipulated regulations, a pay order will be issued. If it doesn't, it is considered as a rejected order (the same process is executed by the superintended APAO/Deputy PAO).

If the bill is passed, it will be approved by the approving authority, APAO (Assistant Pay and Accounts Officer) Deputy PAO. If the bill amount exceeds Rs. 5 Lakhs and up to 1 crore, PAO's approval is

necessary. Bills related to arrears will be sent to the APAO/Deputy PAO. Bills related to Electricity, Water, Telephones, etc will approved by the Superintendent.

After the approval of any bill, it goes to the cheque writing section and then goes to the cheque delivery counter. The DDO or its representative can collect the cheque from the cheque, Delivery Counter.

Based on the nature of claim, PAO issues payment in following manner:

- a. In case of corporations, PAO sends a Letter of Credit (LoC) to the Corporation (e.g. Government Funds allotment).
- b. In case of DDOs, PAO issues cheque for the total amount on the name of Bank Manager, SBH and list of DDOs. The Bank manager credits amount to respective DDOs.
- c. In case of the individual having account in any bank the cheque on total amount will be sent to Bank Manager, SBH and the list is sent to RBI as well. Banks in turn credit amount to individual accounts.
- d. In case of any firm, which has supplied the material (e.g. computer hardware) to the concerned government departments, PAO issues a cheque on the firm name but the cheque will be sent to the concerned DDO.

PAO maintains the list of payments done on that particular day and will assign a number against each issued cheque. This number is called voucher number. From then onwards the voucher references the bill number. All the vouchers are sent to the computer section for data entry. Reports are generated Daily/Monthly and are sent to different departments.

Functions of E-Khazana

National Informatics Centre, Hyderabad has developed Treasury application e-Khazana, " an advanced, integrated software package with the following features:

Outline of the Functions of E-Khazana

The functions of e-Khazana are:

- Token issuing
- Budget verification and monitoring.
- DDO/HOA access verification
- Bill auditing
- Bill process log for entire bill life cycle.
- Pay order generation
- Bank list processing
- Bill status
- Reconciliation with bank scroll

- Daily and monthly reports generation
- Reports on Receipt and expenditure Status at any point of time.

Outline of E-Khazana System Architecture

Figure gives a brief outline of the e-khazana system architecture.

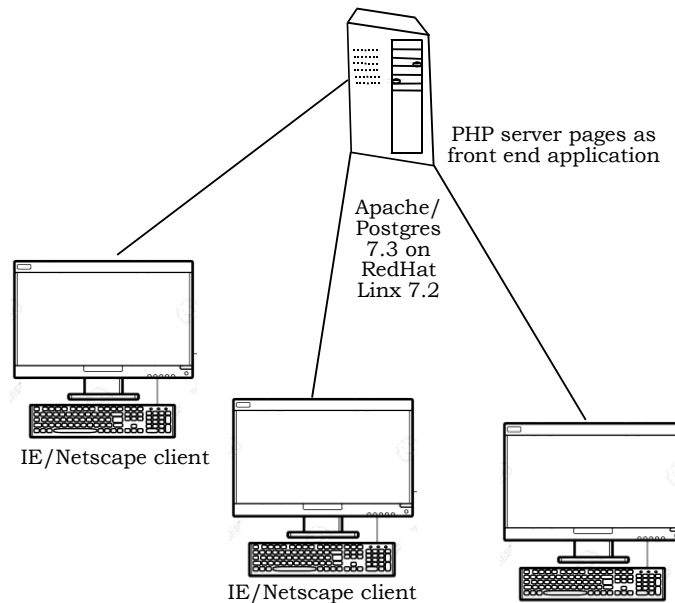


Figure: Illustration of E-Khazana architecture.

Application architecture

It is a 3-tire architecture. Application is built using Hyper Text Pre-processor server pages, HTML and JavaScript. Back-end in postages SQL 7.3.

Automatic of functionality

It is a workflow automation tool used for disbursing and accepting funds to various government originations after auditing the application and documenters for meeting the requirements. The workflow starts with issuing of a transactions ID. Budget and other master data are maintained separately. Check list for each kind of transactions is also maintained as part of master data.

Application flow

Figure illustrates the application flow.

- **SLO:** (Substantiative Level and Officer) process is to maintain cadre strength designation–wise. It should match with the bill submitted by the DDO.
- **Accounts:** Process is for generating monthly accounts for submission to Account General.

Office of Director General for Foreign Trade (DGFT)

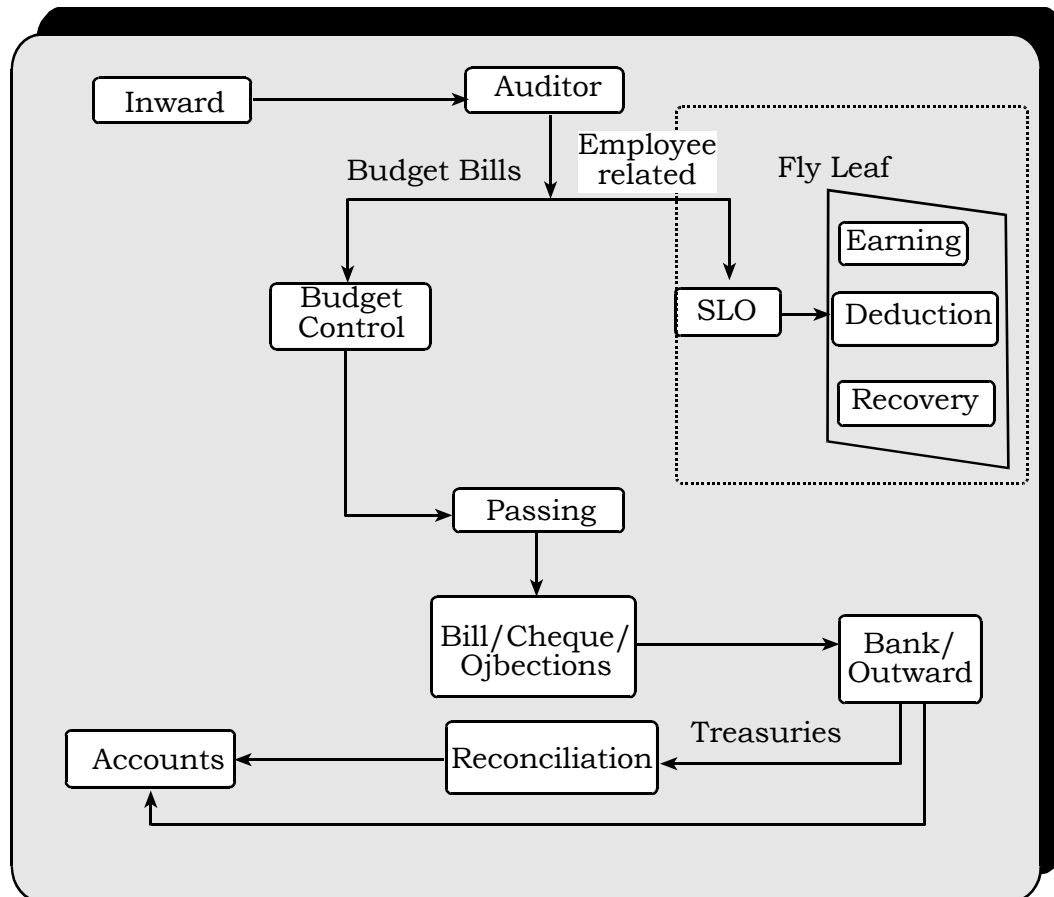


Figure: E-Governance in the Office of Director General for Foreign Trade (DGFT)

Introduction

The era of digital economy is evolving with e-governance being introduced into all economic and trade related departments of the Central Government in India. In this direction the Office of Director General of Foreign Trade (DGFT) * under the Ministry of Commerce is a key player in promotion of Foreign Trade in India.

In the recent efforts to modernize and speed up the processes in the Offices of DGFT located throughout the country, especially its port offices, computerization was introduced with the help of National Informatics Centre (NIC). This ushered e-governance, including digital signatures, in DGFT offices throughout the country.

The EXIM Policy (The Export-Import Policy) of Government of India outlined the goal of computerization processes as to reduce the physical interface and bring in transparency in various

* We thank DGFT for providing the information in this case study.

activities related to exports, electronic data interchange (EDI), and ensure clearance of electronically received application within 24 hours.

Role of DGFT in implementation of Exim policy

The Government of India may appoint any person as Director General of Foreign Trade for the purpose of Foreign Trade development and Regulation (FTDR) 1992 Act. The Director General shall advise the Central Government in the formulation of the export and import policy and shall be responsible for carrying out that policy.

The principal objectives will be met with the coordinated efforts of the state governments and all the departments of the Government of India in general and the Ministry of Commerce and Industry and directorate General of Foreign Trade and its network of Regional Offices in particular, with a shared vision and commitment and in the best spirit of facilitation, in the interest of promotion of trade in goods and services.

The Office of the Director General of Foreign Trade, a Central Government Department is located at New Delhi. It has four Zonal Joint Directorate General of Foreign Trade (ZJDGFT) officers in Delhi (North/CLA), Mumbai (Eastern), Kolkata (Western) and Chennai (Southern), and other 28 port officers across the country.

To know the details of **DGFT** functions, various licensing schemes, etc., please refer to Annexure I at the end of the chapter.

Pre-computerizations scenario: Manual Application Processing Procedure

Normally an exporter should process and Importer Exporter Code (IEC) in order to avail the benefits/exemptions under various licensing schemes, unless specifically exempted. Allotment of fresh IEC numbers, change in the details of the firms are normally done by local offices on submission of application with necessary documents. After obtaining the IEC Number, the exporter can avail any of the benefits of existing licensing schemes mentioning the IEC details.

Once the application is submitted, Receipts and Issues Counter, after scrutiny, allots a file number and issues and acknowledgement to exporter and the application is forwarded to the concerned section. The application is further processed by the concerned section. The application is further processed by the concerned section which prepares a draft note and sends it to the concerned officer. Depending on the decision of the concerned officer, a rejection letter/licence/deficiency letter is prepared and passed on to R and I section which in turn sends the document to the exporter either by post or by hand.

In the process, the application may fall into one of the categories and may be referred to one of the section. In some situations, for example, an exporter obtained IEC in one DGFT port office and having branch office in another port office and the exporter files application in the port where branch office located. The local port office has to get the IEC details from port office where IEC was obtained, and after verification grants the licences for which exporter has applied for. The number of times each life moves between the local officers, the processing time of application and the number of file each dealing assistant has to handle, type-written licences, hand written draft notes to be put up on the file to the concerned officer, manually appending the remarks of the officer, etc. – all this caused enormous delays

in processing the application, any giving the licence to the exporter. An exporter has to visit DGFT office many times to file application, to know the status of his application and finally to get the licence. At Customs Department end, they cannot give clearance to import or export the goods if they do not have IEC. Depending on the category of exporter his application has to be processed locally or at zonal office or it has to be referred back to DGFT.

Moreover, to complete the information required by DGFT/RBIO/Audit Party/Customs and locating the required information from thousands of files (Since inception of the department) is very much time consuming and laborious for DGFT office staff. Mode of transmission of data is through telephone, telegram or through post.

If we consolidate the above, the sequence of activities which were manually performed were as follows:

- Step 1 : Allotment of file number
- Step 2 : Giving written acknowledgement to the exporter
- Step 3 : In-ward/out ward register
- Step 4 : Putting up draft-note
- Step 5 : Appending remarks by officer
- Step 6 : Referring earlier files for any clarification
- Step 7 : preparing rejection letter/deficiency letter
- Step 8 : Preparation of draft licence
- Step 9 : Preparation of final licence
- Step 10 : Obtaining information from other ports (Whenever required)
- Step 11 : Compiling fee collected (Daily/Monthly/Annually)
- Step 12 : Compiling reports required by various agencies and the list goes on.

The manual system was not exporter friendly in terms of time delays caused. Staff of the DGFT offices were on their toes to verify the details of application for correctness and were deeply involved in the decision making process.

Details of Computerization in DGFTs

The National Information Centre (NIC) under the Ministry of Communications and Information Technology, Government of India, was given the task of computerization in DGFT offices during 1990s, in order to have a reliable Information System to cater to the needs of DGFT, customs and related departments and to be more closer to the Trade Community by way of providing the services at greater pace.

Objectives of Computerization

The objective of computerization were to:

- Reduce physical interface-(through Electronic Transactions)

- Monitor and control-both online
- Reduce paper work
- Increase the speed of transactions and to make them transparent while keeping them secure.
- Develop integrated database to help the decision makes.
- Assist in policy making and evaluation.
- Extend quick and transparent services to exporter under various schemes include in the EXIM Policy
- Assist exporter to reduce transaction time and costs and eliminate intermediaries.
- Strive to provide value added services to exporters through information, guidance and hand holding
- Leverage information and communication technology, in-house expertise and coordination with other agencies to achieve the above objectives.

These objectives were sought to be achieved by:

- Internal computerization of organization unit (LAN)
- Setting up connectivity between offices of DGFT (WAN)
- Electronic Data Interchange (EDI) with Trad and Industry, and other government agencies like customs, bank etc.
- Paperless operations.

The following benefits were expected for exporters:

- Single format for licence application
- Instant status of application
- Electronic filing of applications will reduce the visits to DGFT office
- Single web site for all export related information.

The following benefits were expected for DGFT:

- Monitoring of each receipt including licence applications will ensure adherence to time limits.
- Automatic Data Integration for Management information System and EXIM Bulletin.
- Export Obligation Monitoring Automatic Generation of Lists
- Reduction in paper work
- Faster communication between offices and with the outside world
- Easier detection of potential frauds/misuse of licence.

Computerization-First Phase

During the first phase of computerization, comprehensive software package was developed to computerize the Receipts and Issues Counter, Importer Exporter Code Section, Advance Licensing, Export Promotion Capital Goods Scheme (EPCGS), Duty Exemption Pass Book (DePB) and Special Import Licensee (SIL).

Initially, a Pentium system with Unix Operating system, with sufficient number of printer and dump terminals were installed in all Regional/Joint Directorate General of Foreign Trade Offices, Unix/Unity RDBMS was provided. Later Oracle 7.3 was also provided.

Implementation of first phase computerization and thereafter providing Internet facility at all the DGFT ports of the country, resulted in usage of computer generated draft notes, issuance of computerized licences and post-licence monitoring. Due to presence of NICNET connectivity, e-mail facility and internet browsing, communication/information exchange among DGFT offices across the country became easier and much faster.

Over a period of time the large of licensing details was used for statistical analysis and framing/amending the policy.

As outlined in the EXIM Policy to speed up the transactions, reduce physical interface and to bring about transparency in various activities related to exports, to promote electronic data interchange and to clear applications received electronically within 24 hours, NIC and DGFT has come out with an appropriate computerization Plan.

Computerization-Second Phase

With the successful implementation of phase-1 computerization a web enabled LEMIS (Licensing, e-Commerce, Monitoring, Information System) Using Windows NT 4.0/DB2 UDB 5.0 JDK 1.15/was implemented, NIC has established local Area Network (LAN), installed powerful database servers, sufficient number of clients and ink jet/laser printers, UPSs in all port offices. Through RF-Link/VSNL links it has wired all DGFTs offices into a Nationwide Network.

LEMIS (Licensing, e-Commerce, Monitoring Information System) is a web-enabled integrated software package for the office of the Directorate General of Foreign Trade (DGFT), intended to meet the diverse needs of DGFT organization, such as electronic commerce, processing and monitoring of licensing schemes at the national level.

The system was developed in object oriented programming (OOP) language using IBM's DB2 as back-end database under Window NT operating environment. The project team set the following objectives while designing LEMIS:

- Integration of all schemes and to eliminate redundant information
- Easy to use and understand GUI (Graphical User Interface) based interface
- Zero maintenance (maintenance free) solution for user workstations (client) and near zero maintenance (minimal maintenance) at server level.

- Single point entry for all schemes (systems) and easy navigation from one scheme to other with valid access permissions only)

The project team has collected the exhaustive list of user requirements of DGFT organizations as well as Trade and Industry the changes desired in the existing system, then translated these requirements into high level data model document.

Once a high-level data model was designed, the team emphasized on developing common code, i.e. class libraries/functions in order to take full advantage of code re-usability, a major OOP feature. To maintain the uniformity of the system, all the modules have the same appearance and use.

Post Computerization Scenario

Improving administrative efficiency/service of trade by adopting IT in day-to-day functioning, is one of the objectives of computerization. It can be done by the following.

1. E-filing of applications and E-commerce

Exporters now can file application electronically (online/offline modes). Within a day's time the application get processed in local DGFT office and licence is issued if details/documents are found correct. Only the concerned has to come to DGFT with necessary document and get his/her licence. Now DGFT is digital signature enabled; exporter can access and print licence at his end, and he needs to visit the office only once, that is, for submitting the necessary documents.

Apart from e-filing facility, DGFT has integrated the facility of digital signatures and electronic payment of licence fee in the online Electronic Data Interchange Module. DGFT has entered into an MOU with the State Bank of India, ICICI Bank, IDBI Bank, HDFC Bank and UTI Bank for Electronic Fund Transfer facility.

DGFT has tied up with safecrypt Limited for Issuance of Digital Certificate Services for filing of online application for licence by EXIM community. Safecrypt has provided "Safe Exim" Product package (which has already been integrated with LEMIS) for digitally signing and filing the online applications.

2. Electronic Approval on Licensing Authority

LEMIS software has developed in such a way that the competent authority gets access to the application through his computer and approve the printing of licence by clicking approval button. Prior to this, dealing assistant has to generate office note, draft licence from computer, whereas now his efforts are reduced. This also amounts for automation in DGFT offices. Rejection/Deficiency letters are generated in the computer. History of applications, like how many times it is updated/accessed any by whom are available in the database.

3. Importer-Exporter Code (IEC) and Binary Identification Number (BIN)

Every exporter has to obtain an Importer-Exporter Code Number in order to import/export any items under various licensing schemes. For fresh IEC or for any modifications of the firm details one has to apply local DGFT office. As soon as the details are fed/modifications are done, a script (java thread program) is invoked local DGFT computer and details are sent to DGFT server through NICNET. DGFT Server, which in turn a Bin is allocated by computer, Customer will

access such IECs information through DGFT web site and permit import/export of the items/goods. On the exporter's part also, one can view the IEC application details and know whether BIN number is generated or not and accordingly they can approach Customs Department. Here DGFT is rendering service to export/trade community/Customs Department Electronically/through internet.

From 1st April, DGFT is issuing a single IEC against a single Income Tax PAN for all the category of applications. Within this procedure, an exporter having different branches across the country can use same IEC number for his/import/export activity, which was not the case earlier.

Now it is mandatory on the part of the exporter to file annual trade returns online.

4. Getting information for IECs registered other ports

Sometimes it may be required that local DGFT office has to download IEC details registered elsewhere any other port office in the country. This situation arises if branch office/factory is located in the jurisdiction of local DGFT and has applied for a licence for a particular scheme. Through Administration module of LEMIS request of IEC details are registered in the database and once 'e-commerce' script on the local is run, the required information is download to the local server from other port database server.

5. Licensing information at the disposal of DGFT

DGFT server, on a regular basis, gets details of licence issued at various port offices in the country through NICNET for further statistical analysis and dissemination.

Computerization also helps in providing information speedily to export community/trade. NIC-DGFT has hosted a web site on DGIFT server and renders the following service.

- (E-filing of application for DEPB-Post, Advance Licence, DFRC, GEMREP, EPCG) through online/offline modes
- Online help to e-filing (online/offline) for all existing schemes
- Viewing IEC details at exporter's end/customs end
- To know IEC status/BIN number at Customs
- To get a digital certificate online
- Online filing of annual trade returns
- Uploading BRC data (for bank only)
- Online help for filing applications digitally/electronically
- Online check for Deletion Status in case of blacklisting (If a firm is blacklisted, the details are entered on local DGFT server, which in turn is transmitted to DGFT server at New Delhi on daily basis from all ports, in the country. Any exporter can check whether his firm is blacklisted or not, through DEL status option of DGFT web site.
- Information on various Exim facilitation measures, Agri_Export Zones, Certificate of Origin, Impact of Sensitive Items, Medium Terms Export Strategy, India's Trade.

- Export Import Data Bank, ITC (HS) Query Based information and other useful statistical reports.
- Details of EXIM policy
- EXIM procedures
- Public notice
- Notification
- Circulars
- E-mail IDS
- Download which include EXIM policy text, EXIM procedures, DEPB rates, Products covered under SION and IT CHS, and Appendices (Various application formalis).
- Links to web sites of other ports, Viz. Mumbai, Chennai, Kolkata and Hyderabad.

Web sites hosted by Mumbai, Chennai, Kolkata and Hyderabad enabling exporters to know the status of application filed. General information's like, contact whom for what etc. Also, phone numbers and e-mail IDs are provided for the benefit of exporters so that they can mail, or contact the concerned to resolve their problem.

E-way of installing new versions of software at operational offices. The project team is providing latest versions of LEMS on <http://dgftcom.nic.in.dgftdownloads.html>. Local NIC officers/System Administrators of DGFT can easily download and can incorporate new changes without any difficulty. This has really reduced the cost and time of NIC projects team/implementation team, involved in travelling to different operational office to minimal.

Conclusion

In this case study, we have provided details of the efforts being made in the office of Directorate General of Foreign Trade (DGFT) for computerization of web enabled services for citizens. These services include import/export licensing services and also information services and e-commerce services of Indian citizens involved in import/export activities.

PRAJA

Introduction

Researching people in the rural areas, Praja is a concept of providing all government and allied services to the people in the district and mandal (for Block) level. This unique experiment taken up for the first time in West Godavari District of Andhra Pradesh was coordinated by the District Collector, * West Godavari, technically supported by National Informatics Centre, West Godavari District.

The Project Praja (The Citizens) is an effort to bring government close to the people empowering them through IT. In other words, it makes government more accessible to the people. It is an effort to delivery

* We thank the District Collector for providing this information for this case study.

various G2C and C2C services to the citizens in rural areas. The project opens web-enabled rural kiosks termed Praja Seva Kendrams at the mandal level and also at village level. The fully computerized Praja Seva Kendrams are on a district wide network connected through the dial up circuits and interact with the district server acting as a remote access server. This project has been implemented in West Godavari District of Andhra Pradesh by the District Collector. The technical responsibility in terms of network support and all software development, support and coordination has been provided by National Informatics Centre (NIC), West Godavari District Centre, Eluru, Andhra Pradesh.

The Praja Seva Kendram runs a district portal Centre, Eluru, Andhra Pradesh, citizen centric services. These services range from the issuance of various certificates to getting information about various programs and also go to the extent of networking citizens to each other and allow them the flexibility and extent of networking citizens to each other and allow them the flexibility and convenience of mutually beneficial transactions. The horizontal portal is eventually put on the global World Wide Web and thus it allows vertical integration with the expanding frontiers of universal knowledge and information bank. The project allows access to hitherto marginalized communities and therefore helps bridge the existing information gaps and is a step towards digital unit. One unique thing about this digital unite experiment is that many of these Kendrams are run by the women self-help groups, a movement that has been gaining grounds by the day in this district. Many of the Praja Seva Kendrams are run as self-employment units and are manned by unemployed youth from CMEY¹ and PMGRY² beneficiary groups. The project would thus provide Information Leadership to these groups and would help them act as change agents and would make it possible for them to grow in strength and stature along with the project. As a result, the self-help groups and strength and stature along with the project. As a result, the self-help groups and CMEY and PMGRY would draw strength from the project while they would buttress the project with their existing strength, a win-win situation for the twosome.

There is also a myth under circulation that the IT solution is anti-poor and is an unaffordable luxury to nations like India. The digital divide is said to be inevitable. Once has to only realize that IT is only a tool that influence the objective and if the objectives is to make the system more transparent, open and accountable, there is no way it can be anti-poor. Most of the rural areas suffer on account of lack of right information regarding the markets, products, agriculture, health, weather, education, etc. and if all this can rural areas suffer on account of lack right information regarding the markets, products, agriculture, health, weather, education, etc and if all this can be addressed though connectivity and IT, a sea change can be brought in the conditions of the rural communities. This is why creation of a knowledge and information economy can bring in more opportunities and thereby prosperity to the impoverished areas. The project recognizes this fact and envisages that all the village can become knowledge hubs and gain symbiotically from each other and derive benefits from the global networks.

Services Offered

The kind of service offered through this project are described now.

¹ Chief Ministry's Employment Plan.

² Prime Minister's Rural Employment Plan.

Online Filing of Complaints and Grievance

Governments enter into daily of the common citizens in rural areas in many ways. There is a fair degree of discontent amount the citizens about the delivery of these services. The craving for better services couldn't have been delivery of these services. The Craving for better services couldn't have been stronger. Everybody wants an outlet to report their individual or communal grievance and would be very happy if the same is attended to with alacrity and promptness. Project Praja allows citizens to file their grievance with their centers. Every grievance is acknowledged and transferred online to the right office for bringing in prompt field level action.

Grievance have been territorially and functionally categorized allowing case of registration. Every grievance is given a unique number (year, mandal code and serial number) for easy retrieval to monitor and do know the status.

As the grievance is uniquely numbered and recorded in real time in the central database, there is no way it can miss the attention of the concerned officials. Registered grievances can be easily monitored by concerned departments by logging on using their user id and passwords specifically assigned to them for this purpose.

Every grievance can be monitored at the highest level, leading to prompt and effective redressal. Real-time summary statistics and performance summary statement of the individual department can be seen and verified. The citizens can verify and track the status of its disposal online.

The project works on the principle that the citizens need not go to the authorities personally if someone can carry his grievance, and in this case if a telephone wire can do that, where is the need for them to act otherwise? The citizens now need not wait if there is a drinking water problem or a nonfunctioning fair price shop or a government functionary not doing his duty. All this is only one click away now.

Online Application Registration

There are host of area where the citizens approach various their and areas of governments for getting benefits out of various government program. These areas range from getting a loan under self-employment schemes to applying for an old age pension or asking for subsidized agricultural inputs. Instead of moving from offices to offices and getting harsh responses, the citizens just need to come to the Praja kiosk and apply online. They get their acknowledgment number and the rest is taken care of. The module provides for online forwarding, transmission, handling and disposal of such requests, and would therefore minimize the disposal time and the concomitant citizens effort to get their cases redressed. The interconnectivity and linkage with the citizen database would also help in weeding out bogus and repeat cases.

Issuance of certificates

There are many kinds of certificates that the citizens need from the governments, the important among them is the integrated caste, nativity and income certificate. The citizen need not go to the integrated caste, nativity and income certificate. The citizen need not go to the Mandal Revenue Office, but can apply directly at the Praja Kendram, from where the request is transmitted online; the certificates would be prepared and would be made available to the applicant at the kiosk itself without any inconvenience and without the drudgery of shuttling through offices. The project would also provide for the issuance of

death and births certificates at panchayat level (with the implementation of e-panchayat) and also at MRO office.

The philosophy behind this intervention is that the citizen always loathes approaching a government department for the fear of getting discourteous treatment and being subjected to corrupt practices. The project therefore improves upon this interface and expects them to come to a center run by their own peers, a place that is much more accountable, open, transparent and subject to public scrutiny. This will automatically mean spending some nominal amount of money (Rs. 56 or Rs. 10) for getting the work done from the Praja Centre in the Village/Mandal so as to compensate for the self help group members or PMGRY/CMEY beneficiary

Issuance of Land Records

The plan document of the Seventh Five Year Plan rightly observed, "Land records form the base for all land reforms and therefore regular periodic updating of land records is essential in all states."

The information relating to land plays a very important role. These may include primary information about land presented in terms of its geological information like the shape, size, land forms, soils; economic information related to land use, irrigation and crops; and the information pertaining to the legal rights, registration and taxation.

The manual system of land records maintenance was highly opaque wherein the land records were maintained by the village accountants and therefore there was a situation of virtual monopoly of village accountants over these records. Records were not open to public scrutiny and were update many and extortion for not only provision of land records to them from the village official but also for processing request for change in land title.

On many occasions such delay was unintentional. Even if a village accountant was willing to give such records in time, he was not available when farmers wanted him most as he was manning more than four to five villages. Therefore, there was no certainty about timely availability of such records when a farmer required them.

Land records contain various useful data such as soil type, irrigation details, trees, crops grown, crop yield etc. All such data are very valuable for various administrative purposes. As data were previously maintained manually, it was not possible to collect and analyze such data with the result that mine of such useful data was not used in any meaningful way. However, keeping this in view, the land records in the state have been computerized and all the Mandal revenue office now maintain these records.

But despite the availability of computerized database, no public outreach to this database was put into place. Further validation of data was also not properly done leading to errors in database. The printed records were distributed to the farmers with proper validation and the updating or any use of the database was mostly forgotten.

The process of mutation (Change in land title) was also very cumbersome. Applications were being given to Mandal/office village official who virtually enjoyed discretion of either processing them or not. As records are maintained in a decentralized manner, there is no reporting mechanism available at the district level about the pendency of such applications. Lack of any monitoring mechanism in the manual system makes farmers amenable to all pressure from the hierarchy of the department.

The project Praja envisage putting this data into the public domain to support development of citizen centric land records system . The result would be evolution and a transparent and effective land record delivery system which would fully address the insecurities and concerns of the farmers.

The farmers can approach the Praja Kiosks in the village itself and ask for the land records abstract for the Fasli for which Jamabandi has been done and get that instantly. In these kiosks farmers can see their land related information without any intervention or help. They can also apply for t6he latest extracts and their request can be processed online by a connection to the Mandal Revenue Office and the land record extract can be produced and offered as a print out to the owner of the land who requested for it. The monitoring of such requests and their disposal at higher levels would ensure timely compliance and would greatly benefit the citizens.

Besides these, they can also lodge request for mutation to their land records (change in land title) at the computerized kiosks, get acknowledgements for the same and can monitor the process at the Praja kendrams themselves without going to MRO Office. They would then get their updated land record in fixed time frame without the need of approaching any authority. As against earlier time of 70-200 days, mutation would now require less than 30 days, that too without going to any government office. It is also envisaged to have a fully online system to carry out mutations on land records data and the manual changes without moving through the database are proposed to be banned.

Farmers can also get the official status report of their request for mutation which would let them would let them know the stage at which their request is pending. This status report would help them in enforcing their right of getting the record mutated in the prescribed time.

The entire Land Records Management Information System (LRMIS) software (similar to BHOOMI) software of Karnataka) has been developed by NIC, Andhra Pradesh State Unit, Hyderabad.

It is also proposed to network the SROs with the mandal revenue offices so that the changes in title taking place are incorporated in the village accounts within the stipulated time without fail. The necessary software has already been developed by NIC, Andhra Pradesh State Uint, Hydrabad and Implementation of this software is under progress.

Online Auctions and Biddings

One fundamental reasons for the rural- urban divide is the lack of well developed market in the rural areas hindering efficient sale and purchases of goods, As a result most of the decision made by the farmers and the rural poor are based on insufficient information and are therefore sub-optimal, and in majority of the cases go against them. The middlemen present in all such areas therefore are having a field day and the citizens suffer. The project envisages filling in these information gaps and allow the citizens enjoy the facility of posting their products, for online auctions at any of the Praja kendrams. They have the convenience of posting their product details along with the specification and the expected price. As the information is shared by the other mandal kiosks, by online bidding there is a possibility of giving a beneficial situation to both of them. As the portal is vertically integrated through internet, online queries from other parts can improve the available choices.

The kind of product that can be put for auction may include agricultural commodities, farm implements, land or buildings-literally anything that a rural household would like buy or sell in the rural areas. This also opens the possibility for the DWACRA to citizens horizontally within the district and vertically

outside. In short, as days go by and as the citizens get used to this facility, it may trigger a chain reaction for opening up the window of opportunities to rural populace to have more information choices in their trades. In other words, this will enable rural e-commerce.

Online Mandle (Market) Rates

Ideally an average household makes a decision before buying or selling the goods and products based on the information they collect locally. The advent of communication facilities like telephone, television, etc. have widened their information base. Nonetheless it would have been much better if they had an easy access to the rates prevailing as on that moment in various PACS or rythu bazaars and other mandis within the district and elsewhere. The project Praja envisages doing that right.

Every kendram will upload the market rate prevailing in their area which would then be available to everybody else. The rates in other market elsewhere in the state would also be available for the citizens to watch the trends and make the right decision after weighting all options. This can be achieved by using the portal on agricultural markets maintained by Marketing Department of Government of Andhra Pradesh, with technical support by NIC, Andhra Pradesh State Unit, Hyderabad.

Payment of Dues

The citizens requires a friendlier environment to pay their dues, be it the payment of electricity bills or the telephone bill or the land revenue. The Praja Kendrams envisage offering this facility to the citizens so that while transacting other services they can clear off their pending accounts as well.

The individual departments have the benefit of expeditious collections and also save on collection costs and at the same time also monitor their collection performance. In future, payments of private organizations like cellular phone can also be dovetailed.

Easy Access to Information

The right to information has now been recognized by the parliament, Praja kendrams through the District portal would allow access to all kinds of value information to the citizens not just with respect to the government programs but also would allow access to the databases like old age pensions, ration cards, multipurpose household survey records, beneficiaries under various self-employment schemes, children database, civil works and so on. This would induce transparency in the implementation of these programs and would facilitate weeding out ineligible cases so that corresponding benefits can be passed down the needy. Efforts in this direction are being taken by National Informatics Centre, Andhra Pradesh State Unit, Hyderabad by hosting online databases on the above mentioned topics.

The project recognize that no matter what rhetoric is, real right to information is possible only if the information is put into the public domain and new technologies provides the most cost effective solution for doing that.

Matrimonial Services

The networking of citizens with each other for mutually beneficial transactions should be the ultimate objective for a vibrant civil social. Other than the government-citizen service, host of citizen-to-citizen (C2C) service are proposed to be rendered through this project.

As a part of that, a marriage bureau would be operationalized so that prospective brides/grooms can place their bio-data eliciting suitable offers and gradually as things move along, a huge record of eligible brides/grooms is created making search for life partners easier and cost effective. The package designed would also keep matching points so that the search is quick, accurate and fast.

Online Civil Supplies Allotment

The Public Distribution System through a string of fair price shops (FPS) provides access to the citizens to many essential commodities. The project proposes to bring the card holders' list into possible domain to so as to weed out ineligible citizens.

There is also a need to prevent frequent movement of FP dealers to various offices. The facility for online stock entry to the dealers through the kendram is being provided and they also would have the ease to collect allotment order online from the Praja center without wadding through the civil supplies offices and greasing palms at various tables.

Tele Medicine

Provision of basic and primary health care is the principal duty of a welfare state. There are, however, many situations when expert advice is needed and instead of the patient travelling all over, the request can travel on wire and the right prescription can be obtained in return.

This can also minimize citizens' dependence on the hospitals, and their falling prey to quacks. Soon, the physical barriers of distance would hold no meaning. Also, the gradual linking with other health sites would improve things further.

Tele Agriculture

The district, being predominantly agriculture based, generates lot of questions related to farm practice which many times go unanswered, resulting in incorrect input applications. The portal would act as round-the clock helping for handling such queries. The only thing the farmer need to do is to come to the kendram and place his question for an answer from the experts. Some of these are million dollar questions and a timely redressal can impinge greatly on the productivity.

Consumables Management

The management and procurement of consumable required by various wings of administration including the primary health centers is being provided through the Praja Kendrams. This also would have a bearing on reducing the procurement costs by effecting economies of scale. The cost of procurement would also be available for public scrutiny so as to make it more transparent and accountable.

Forms Download

There are various forms that every department has for use by the citizens. The Praja Kendram would act as a one-stop shop for downloading all such forms. Any changes in them would also get reflected in real time.

The Praja kendram, through its portal, expects to provide a virtual meeting place for the citizens to discuss issues relating to the district/villages. Its problems and prospective solution. The citizens can now freely interact with each other and post their ideas. This would act as an online forum for them to

ventilate their grievances, air their opinions and cause necessary social changes. It also provides opportunity to conduct opinion polls on important topical issues leading to improved decision making.

E-Education

The digitized inputs on computer enabled education would also be made available in due course so that the analytical and thinking abilities of rural students are enriched opening to them the advanced frontiers of knowledge's. Internet, being the worlds' biggest library, would also help the citizens gain access to a new world of information.

Common Accounts Keeper for Self-Help Groups

The Praja kendram, which is run by the women self-help groups, would provide a virtual meeting place and a focal point for synergizing and pooling in their efforts. Instead of individual groups maintaining their own accounts, they would be in a position to use the Praja Computer for maintaining their internal lending records and also to enter into online transactions with their banks. Some of the banks also propose to install all ATM in some kendrams which would further facilitate the groups.

Message Board

The portal, through the kiosks, would enable the administration to pass down important social communications and advocacies for broadcast to the communities. Private advertisers can also use this medium for popularizing their brands. The citizens can also make use of this facility for sending quick information to their near and dear and the SHGs running these kiosks can literally act as a courier service with minimal effort.

The list can become even lengthier as the areas where technology can make governments do their job better are endless and as the days go along project would keep evolving with innovative ideas.

To say the least, if we fail to act now, the information gap risks may widen into an unbridgeable gulf that may increase global inequality and leave the poor further behind. But if we approach the matter with the same kind of urgency and application as the commercial ".dot.dom" sector, then we have every chance of building a strong, new wired future that not only includes the world's poor but gives them an unprecedented opportunity to left themselves out of poverty.

Additional Services in Phase II

Following are the services being added in Phase II as a part of Village Panchayat Function:

- (i) House/Property tax. The citizens will be able to pay House/Property tax to the Village Panchayat in this Service.
- (ii) Trade licences. Citizens will be able to obtain trade licences in this service.
- (iii) Works monitoring. The rural citizen can make proposals for certain works such as roads, school buildings, culverts, brings, etc. to be carried out in the village to the village Panchayat who will approve it in Gram Sabhas and pass resolution to the effect. Such works are taken up by Engineering Department by assigning contracts. The progress of such works can be monitored by this software.

All the above software modules are developed as a part of Village panchayat Information system under e-panchayat project by National Information Centre, Andhra Pradesh State Unit, Hyderabad.

Conclusion

In this case study on Praja Rural E-Seva Project of West Godavari District of Andhra Pradesh, we learnt about the various services from the Government offices that can be carried to the people's by people participation. Even though this means additional cost to the people, it is still a successful service especially in terms of its ease of accountability and utility for common people especially in terms of its ease of accountability and utility for common people in the villages. In addition to providing government services this project encourages rural e-commerce and rural cyber forms in the villages. This project has been emulated by other districts of Andhra Pradesh and a Statewide rollout plan for 6000 Rural Service Delivery Points (RSDPs) by connecting the telephone booths into Kiosks is in the pipeline.

E-Seva

Introduction

e-Seva in Andhra Pradesh, Friends in Kerala, Sukhmani centres in Punjab, the Lok-Mitra project in Rajasthan, Coimbatore-Online in Tamil Nadu and Bangalore One in Karnataka are some of the pioneering G2C initiatives in India. E-Seva, one of the e-governance initiatives, offers a wide spectrum of citizen friendly service to save citizens the trouble of running around various departments.

e-Seva is, perhaps, the best example and one of the earliest attempts to take the benefits of e-governance to citizens. Pioneered by the then IT secretary in Andhra Pradesh, J Satyanarayana, currently CEO of the National Institute of Smart Governance, and driven by the then chief minister, Chandrababu Naidu, e-Seva started as the TWINS project to provide multiple citizen-facing services of the government under one umbrella. Instead of citizens having to go from one department to another, multiple government services are provided under a single window. It is particularly of value to less-affluent and less-influential people, who do not enjoy the luxury of servants or peons who can spend time standing in queues.

e-Seva has several innovations to its credit. The people interface issue has been well addressed and the effects of re-engineering are there for all to see. The ambience is good, the place is clean, less noisy and there is an air-conditioned environment, uncluttered by dilapidated furniture. There is access to drinking water, facilities like ATM, internet kiosks and phones things one does not normally associate with government departments in India. The services are offered at dozens of centres, each with several counters, in a public-private-partnership model. The computers, printers and internet access are provided by the private operator, who is paid a fixed service charge for transactions completed at e-Seva counters, with the government providing the data.

Currently, the services include:

Salient Features of E-Seva

Following are the salient features of e-seva:

- Provides real time online transactions.

- Integrated Citizen Service Centres (ICSCs) operates from 8 a.m to 8 p.m. on all working days and is open on Sundays and Second Saturdays also 9.00 a.m. to 3:00 p.m.
- Electronic queuing system and efficient transaction of business reduces waiting time.
- Distribution of centre in all major areas surrounding the city, likely to expand to other parts of the State also.

Implementation Model-Public Private Partnership

This project has been implemented to a Public-Private Partnership model. The technology is based on a 3-tier architecture. The transactions are done on a real-time basis. The servers of different departments are connected to the e-Seva Data Centre, which is in turn connected to different ICSCs. The connectivity is provided through leased lines with ISDN line as backup. Transactions conducted at the ICSCs are recorded directly on the server of the department concerned, after duly according the same in the central server.

Each ICSC has 10-14 computerized manned counter with one PC, printer, bar-code reader per counter to handle the services.

Hardware and Software

- Two sun E250 servers with 100% redundancy and two Compaq ML 530 database servers with 100% redundancy.
- Oracle 9i AS database servers with 100% redundancy.
- Oracle 9i AS application server running on Sun Solaris 8
- Oracle 8i Release 3 database running on window 2000
- On firewall server
- One web applications server for portal
- One web application server for portal
- One network monitoring server running Cisco works on windows 2000
- Java based front-end
- Oracle backend in all departments.
- 10 kVA UPS for the clients in Khairatabad centre with one-hour backup and 5VA UPS all the server in the data centre with 45 minutes backup
- 5 kVA UPS in all centres with 30 minutes backup
- 12 clients and 10 printers, including a laser printer and one PVC and printer for printing driving licences.

Service Centres

At present there are 46 e-Seva centres spread across the twin cities of Hyderabad and Secunderabad and Rana Reddy District, and two bank branches of Andhra Bank and State Bank of Hyderabad, respectively.

Extension to Districts

Regarding the replication of e-sewa in municipalities in the Stage, it was decided to expand the e-Seva services in all 117 municipalities in the state, which cover 85% of the population of the state.

Services Available

(i) Payment of utility bills

- a. Electricity bills
- b. Water and sewerage bills
- c. Telephone bills
- d. Property tax
- e. Filing of CST returns
- f. Filing of A2 returns of APGST
- g. Filing of AA9 returns of APGST
- h. Collection of examination fee, affiliation fee and recognition fee of Board of Intermediate.
- i. Filing of IT returns of salaried class
- j. Property tax of L.B. Nagar municipality
- k. Property tax of Kuktpally municipality
- l. Property tax of Qutbullapur municipality
- m. State of prepaid parking sticker of MCH.

(ii) Certificates

- a. Registration of birth
- b. Registration of death
- c. Issue of birth certificates
- d. Issue of death certification

(iii) Permits/Licences

- a. Renewal of trade licences
- b. Registration of new trade licences
- c. Change of address of a vehicle owner

- d. Transfer of ownership of a vehicle
- e. Issue of learners' licences
- f. Issue of driving licences (non-transport vehicles)
- g. Registration of new vehicles
- h. Registration of new vehicles
- i. Quarterly tax payments of autos
- j. Quarterly tax payments of goods vehicles.
- k. Lifetime tax payments of new vehicles.

(iv) Reservation

- a. Reservation of APSRTC bus tickets
- b. Reservation of water tanker.

(v) Other services

- a. Sale of passport application forms
- b. Filing of passport applications
- c. Sale of non-judicial stamps
- d. Sale of trade licence applications
- e. Sale of national games tickets
- f. Sale of entry tickets for WTA
- g. Sale of EAMC:ET applications.

(vi) Internet services

- a. Internet-enabled electronic payments
- b. Downloading of forms and Government Orders.

(vii) B2C Services

- a. Collection of telephone bill payments of TTL.
- b. Sale of new AirTel Magic cards.
- c. Top up/recharge of Air/Tel Magic cards.
- d. Sale of entry tickets for tollywood star cricket
- e. Sale of entry tickets for cricket match (RWSO)
- f. Filing of Reliance CDMA mobile phone connections
- g. Sale of prepaid India Telephone Cards

- h. Sale of application for Model EAMCET

Additional Service Proposed

1. Railway reservation
2. Sale of movie tickets
3. Payments of traffic related offences
4. payments of degree examination fees of Open University
5. Sale of I-CET applications
6. Online reservation of darshan tickets for Lord Venkateshwars at Tirumals
7. Collection of bill payments of Idea Cellular
8. Collection of bill payments of HUTCH
9. Issue of encumbrance certificate
10. Market value assistance
11. General insurance
12. Reservation of tourism tickets for accommodation
13. Reservation of tourism bus tickets
14. Call center
15. Indian airlines ticket reservation
16. Life insurance premium payment
17. Issue of caste certificate
18. Sale of Indira Vikas patra
19. ATM services
20. Collection of bill payments
21. Renewal of drug licence
22. Issue of bus passes
23. Collection of trade licences of Labour department

Conclusion

In this case study on e-seva of Andhra Pradesh Government, we surveyed the salient features, details of services offered and proposed future services E-Sava is the most successful among all projects in India that have reached common man with essential services of the Government. Municipality and also Utility Companies all under one roof.

E-Panchayat

(Electronic Knowledge Based Panchayat)

Introduction

E-Panchayat is a software product conceptualized, designed and developed by National Information Centre, Hyderabad, Andhra Pradesh, as a part of its E-governance initiatives. E-Panchayat has been designed taking into consideration all the information and knowledge management requirements in a Gram Panchayat. The 1992 Panchayat raj Act of Government of India, and the Success stories of the Gram Panchayats in various states in the country have been taken into consideration while developing e-Panchayat. The Government Order issued by the Government of Andhra Pradesh regarding the village secretariat and the functions of the village secretary besides the elected representatives of the gram Panchayat are all considered in this package. Thus e-Panchayat fits well into gram panchayat are all considered in this package. Thus e-Panchayat fits well into the information systems at gram panchayat level. The e-Panchayat fits well into the information systems at gram panchayat level. The e-Panchayat fits well into the information systems at gram panchayat level. The software is web enabled, and citizen-centric. Therefore, even with less overheads the village level functionaries and the citizens are envisaged to benefit by e-Panchayat.

The Modules

E-Panchayat comprises nearly 30 main modules, and Nearly 150 sub-modules, in line with the 30 sectoral functions of the gram panchayats. Ramachandrapuram Gram Panchayat (Merdak District, Near Hyderabad, Andhra Pradesh) was taken as panchayat for e-Panchayat.

Gram Panchayat Administration

The module facilitates provision of following information services:

- Minutes of meetings
- Schedules of Gram Sabhas
- Agenda update
- Certificates, Licences
- Bill payments
- Attendance monitoring of the Panchayat functionaries
- Gram Panchyat Meeting Management
- Gram Panchayat Cleanliness Monitoring
- Registration of Births/Deaths/Marriage Etc.
- SGSU works monitoring, issue of trade licences
- Pension schemes management system

- Self-help group and other villagers welfare schemes management system.
- Assets management, property tax assessment and management.
- Property lost/found reporting system
- Encroachments checking and monitoring module
- Government orders searching and retrieval system
- Village statistics maintenance and reporting system
- Intensive sanitation management information system.

Agriculture

This module enables the elected representatives of the gram panchayats to manage the farmers' grievances.

It facilitates rendering educational services on the best agricultural practices to enhance the yield and reduce expenditure practices to enhance the yield and reduce expenditure and enhance the quality of produce for the farmers. Also it facilitates its agriculture and related department to provide season-specific, region-specific information services to the farmers, apart from offering of counseling services to the farmers by agriculture experts.

This module also enable the gram Panchayat secretariat to report the agriculture production details of the villages.

Irrigation and Water Conservation

This module provides information on irrigation canals to the public. It facilitate education on various water harvesting and conservation scheme. The module enables reporting of problems on pipelines, canals, etc. and subsequent review of problems by Sarpanch (or the head of Panchayat or Village Council).

This module facilitates provision of the following information services:

- Veterinary counseling services
- Information on animal diseases
- Information on milk procurement and quality management
- Reporting on breed improvement program.
- Livestock data collection and reporting

Fisheries

The fisheries module facilitates provision of the following information services:

- Fisheries, counseling
- Provision of information on potential fishing zones

- Provision of fishermen welfare scheme related information
- Rendering market related information services.
- Weather forecasts.

Social Forestry

This module facilities provision of the following information services.

- Mobilization of volunteers by secretary.
- Submitting report on illegal felling of trees.
- Procurement and management of information on the vegetative resources of the gram panchayat.

Elections

This module provides the following information services:

- Registration of voters
- Objection to voters list
- Elected representative information
- Publication of electoral roll
- Dissemination of electoral roles.

This module extends following information services:

- Loans/schemes for entrepreneurs
- Raw material sources
- Market information
- Marketing services
- Loan repayment
- Information on income generating projects.

Housing

The module provides information service such as:

- Low cost housing technique information
- Apply for housing scheme
- Loan recovery status information
- Information on housing

Water

Information services provided under the module are:

- Information on water clearing technology
- Approved vender's information for water treatment.
- Water sources information
- Problems reporting and counseling

Fodder and Fuel

This module provided information services on:

- Smokeless chullas
- Bio-gas plants
- Application for sanction subsidized loans for setting
- Problems reporting in maintaining the plants
- Request for samplings
- Energy saving techniques.

Roads, Bridges and Culverts

Under this module the information services available are:

- Submission of proposals by people
- Authentication of cost estimates
- Maintenance problems reporting
- Monitoring the repair activities.

Street Lighting and Electric Supply

The module facilitates provision of following information service

- Proposal for erecting polls and transformers.
- Problems reposting on electrical lamps replacement
- Status on bulbs procured/replaced
- Electrical correction application.
- Payment of electrical bills

Alternative Energies

Information services available under the module are:

- Information on Solar Water Heater
- Solar Cookers/Dryers
- Wind Mills. Wind Zoning information

- Information on Bio-gas plants (Rich busk based)
- Submission on available bio-mass.
- Directory of manufacturers

Poverty Eradication

The following information service are provided by this module.

- Authentication of BPL status
- Applying for schemes by citizens
- Verify credentials
- Analyze the requests
- Funds disbursement status
- Feedback on the beneficiaries
- Information on income generating schemes.

Primary Education

This modules provides information services on:

- Inventory of schools
- Resource available at schools
- Teachers information
- Teaching materials
- Enrolment status
- Information dropouts

Technical Education

Information made available under the module are as follows:

- Inventory of ITI and Polytechnic Institutes
- Details of courses offered
- Apply the administration
- Schedule of training program
- Problems reporting and counseling by technicians
- Results and other information.

Library

- This module facilities provision of information services such as

- Information on library tax collected
- Details on expenditure on books procurement in village library
- Requests for new books information by Public
- Preservation of books.

Cultural Program

The module facilities provision on information services such as:

- Directory of libraries
- Information on library tax collected
- Details on expenditure on books procurement in village library
- Requests for new books information by public
- Preservation of books.

Cultural Program

The module provided information on:

- Enrolment of articles
- Training the artist
- Schedule of program
- Information on opportunities

Markets

The modules facilitate information service on:

- Price information on food grains
- Vegetable price
- Marketing services
- Scheme information
- Marketing

Health

This module provide following information:

- Disease information
- Communicable diseases
- Attendance of doctors in PHC
- Status of medicines and consumable

Women and Child Welfare

The modules extends information services such as:

- Immunization schedules
- Nutrition program, ICDS schemes information
- Eligible children and nursing mothers information updating
- Schedule of campus
- Counseling by health workers

Social Welfare

Under the module information service available are:

- Atrocities reporting
- Welfare scheme information
- Inventory handicapped persons
- Donors information
- Counseling
- Apply for schemes.

Cottage Industries

- Inventory
- Counselling
- Loans application
- Marketing services
- Raw materials sources information

Weaker Sections Welfare

This module offers information service on:

- Atrocities reporting
- Welfare scheme
- Inventory and weaker artisans
- Marketing service
- Produce of weaker sections

Public Distribution

Information services available in this module are:

- FB shops
- Stock status
- Card holder information
- Problems report

Protection of Property

The module facilitates information services such as:

- Encroachment complex
- Status of encroachment
- Maintenance
- Tax collection bills.

Land Records module of e-Panchayat enable Land Records Maintenance by village secretary and rendering verification and certification service to the citizens. This also facilitates land revenue management, with appropriate reports to the revenue/gram Panchayat official and the citizens.

Village Accounting System (VAS)

Follows are the features of VAS:

- Cash transactions
- Bill receipts
- DDS and challan receipts
- Automatic challan submission
- Treasury payments
- Classified accounts
- All kinds of taxes, fee collections (online/offline)
- Cheques/DDs reconciliation.
- Reports (Daily collection report)
 - Ersalnama
 - Chilla
 - Monthly classifications
 - Challan receipts
 - Demand and collection reports.

Other Village Services (OVS)

Births and Deaths Registrations

- Capturing of event details
- Name inclusion
- Non-availability certificate
- Registration of events
- Issuance of certificates
- Statistics.

Property Tax and Vacant Land Tax

- Assessment as per monthly rental value
- Warrants and distrains
- Title transfers
- Demand collections
- Write-off and temporary remissions
- Vacancy remissions
- Exemptions
- Special notice/endorsements/demand notices
- Registers/Field books/Defaults

Dangerous and Offensive Trades

- Issue of licenses
- Cancellations
- Renewal certificates
- Demand notices
- Fee collection.

Water Tap Connection and Water Tax Collection

- Sanction of metered/Non-metered connections
- Disconnections/Reconnections
- Regularization
- Work orders/Endorsements
- Demand notices
- Collection of water charges.

Advertisement Tax

- Advertiser registration
- Unauthorized advertisements
- Regularizations
- Renewals
- Tendered advertisements
- Demand notice
- Collection of rents/Lease amounts

Building Permissions

- Building approval for residential/groups housing/now
- Housing/commercial/educational institutions/industries
- Planning fee
- Technical scrutiny
- Proceedings
- Issue of occupancy certificates
- Unauthorized buildings.
- Unauthorized buildings
- Encroachments.

Building Permission

- Building approvals for residential/groups housing/now
- Housing commercial/educational institutions/industries
- Planning fee
- Technical security
- Proceedings
- Issue of occupancy certificates
- Unauthorized buildings
- Encroachments

Projects and Works Monitoring

- Contactor registrations
- Standard schedules rates
- Estimate cost value

- Fund allocation
- Tendered notices
- Tender schedules
- Work order
- Execution of work orders
- Monitoring of works
- Payments and recoveries

Grievance Redressal

- Grievance registrations.
- Pendency monitoring seat/section wise.

Assets and Inventory

- Assets/Inventory registrations
- Maintenance of assets
- Inventory management
- Rental/Lease of assets

Court cases

- Filing suits
- Monitoring of cases at different stages such as admission, interim order, hearing, judgement, appeal, disposal of suits.

Financial Accounting

- Head of accounts
- Payments
- Receipts
- Chitta and each book
- Demand collection balance.

Disbursement of Old Age/Widow and Disable Pensions

- Accepting applications for old age/widow/disabled pensions from anywhere on the internet under NOAP Scheme of Government of India Generation of acknowledgement slip for the same.
- Generation of acknowledgement slip for the same
- Reporting the sanction status of the application accepted

- Online disbursement of old age pension along with generation of acquaintance roll
- Generation of list and summary reports and disbursement and funds details for monitoring the scheme progress
- Reporting the disbursement status of the Beneficiaries.
- Linkage to MPHS data and SSIDs for verifying beneficiaries data.

SGRY and Other Works Monitoring

- Accepting work requests from anywhere on the Internet
- Capturing work details related to all the works as accepted by the Grama-Sabha resolution along with work estimates and technical details
- Capturing tender details of Works funded by Gram Panchayat.
- Capturing work stages and sanction details of works
- Capturing funds details on allocation and releases with respect to cash and food grains as part of SGRY, 11th Finance Commission Schemes
- Allotment of laborers/beneficiaries to works
- Payment of wages in case and food grains to beneficiaries along with generation of acquaintance and maintenance of master rolls.
- Generation of work agreement and work orders to agencies/contractors in case of works funded by gram panchayat
- Capturing voucher-wise works expenditure details
- Reporting of physical and financial progress of works
- Reporting on man-days generated with respect to SGRY stream I and II schemes
- Generation of cash books related to SGRY, EFC and GP funds.
- Reporting to MDO and DPO on Government of India formats
- Generation of beneficiary lists and work stage details on all works
- Reporting of work status.

Conclusion

E-Panchayat aims to cover all information requirement of the village Panchayat, both from the staff angle and the citizen angle.

E-Panchayat is already operationalized (by National Information Centre, Hyderabad) in several pilot village in Andhra Pradesh. A rollout plan for implementation at the national level is being taken up. This case study summarizes the main features of e-Panchayat.

GENERAL INFORMATION SERVICE OF NATIONAL INFORMATICS

Introduction

In this era of information, right to information is vital. To collect data about various walks of life, churning them into useful information is a gigantic task. In view of the availability of NICNET throughout the country including all states and districts and with the advent of Internet, reaching the people with information (and vice versa) is no more a dream. The General Information Service Terminal of National Information Centre (GISTNIC) is a national project, meeting the challenges of data collection, moulding it as useful information and making it available for public access.

Objectives of Gistnic

The objectives of GISTNIC initiated by the National Information Centre are:

1. Offer online general information/database services to common public by deriving useful information from various existing databases and other information sources.
2. Act as a NICNET/Internet based communication channel for the Government to assist in disseminating important development information to common public.
3. Popularize informatics culture into the life style of common public for improving quality of life.

Pre-Internet Era

GISTNIC was available throughout the country through NICNET, the first ever Governmental communications network and at specially designed GISTNIC booths that are strategically located at secretariats, legislative assemblies, libraries, airports, hotels etc offering online information services such as:

- Statistical profile of India.
- District Level Economic Indicators
- District Profiles
- University Education Guide of India
- Tourist Guide of India
- All India Railway Time-Table
- Hotel Guide of India
- General Information Guide
- All India Hospital Directory
- Country Profiles
- Work Economy Tables

- Industrial Directory of India
- Employment Guide
- Investment Guide

GISTNIC was first made available on the NEC-1000 mainframe system over NICNET. Upon invoking GISTNIC a Main Menu was displayed. One can select any one of the displayed service guides. For example, one can select Tourist Guide by selecting the Tourist Guide by number and pressing Return key. Further selections of the sub-means are made to furnish other details. Similarly, all other guides and schedules were made available. Data was being updated every month. At that time GISTNIC was the only source of information in India on the network.

Internet Era

GISTNIC is designed with an objective to make available the general information to public using the state-of-the-art technology. With the advent of Internet and worldwide web, GISTNIC was quickly geared up to cater to the needs of the internet users.

The content and scope of information provided on GISTNIC span a wide range of potential users like farmers, tourists, students, researchers and Government departments. The information requirements of individuals vary widely. The data flows constantly at regular intervals from various sectors such as agriculture, corporate industry, and financial institutions and so on, and the same is being updated regularly on GISTNIC.

In addition to the information being derived out of the databases maintained for government applications, a large amount of published and open information which are noteworthy for the common man is also being provided through this service. GISTNIC project is presently offering information categorized into the following areas:

1. Agriculture statistics
2. Corporate statistics
3. Primary Census Abstract-91
4. District profiles
5. Public finance
6. Education after 10+2
7. GIST-educations
8. GIST-STD codes
9. Rural technologies
10. Exports statistics
11. A.P. Engineering Admissions
12. Social Sector Statistics

13. Industries directory
14. Tourism information
15. Industrial statistics
16. Traditional Sciences
17. Industry Infrastructure
18. Village amenities
19. Primary Census Abstract-1991 in Hindi, Telugu, Tamil
20. Provisional Census -2001 population in A.P. in Telugu
21. Government orders of A.P., M.P., Punjab
22. A.P. Marketing
23. A.P. Vaidya Vidhan Parishad
24. A.P. State museum
25. A.P. Cooperatives
26. A.P. Technical Education
27. Directory of World government web sites.

In addition to the above sectoral information and web sites, links are also provided to various public utility information as follows:

- South Central Railways
- Indian Newspapers
- Indian Government web sites
- Indian Railways
- Train Ticket (PNR) Status
- Trains between Two places
- Indian Airlines
- Cochin Airport
- All India News
- Doordarshan
- India Population Commission

A brief description of some of the services is as follows:

Agriculture Statistics

This module provides statistical information on rainfall, area under various crops, fertilizer consumption, details of credit, irrigated area, trends in utilization, operational holdings and statewide agriculture output i.e. ,area, product and yield of various crops.

Education

This is a comprehensive information portal on educational resources in India and abroad and is powered by up-to-date, reliable information. It is the best resource to the students, educational institutions, academic experts and all others in education field wherein a lot of information on course, degrees, colleges, and professional course-state wise is made available. To know more about the career prospects in IT field after Intermediate, the separate module 'Education after 10+2' is the right choice to browse and know. Information about various venter certified IT programs is also available.

Tourism

India being a paradise for tourists, GISTNIC has provided a lot of information about various tourist spots in India. One can categorically search based on a region (south/north/east/west) historic spots, religious spot etc. or make a state wise search. The data includes how to reach the spot, where to stay, important place to visit around it, climatic conditions and the like Pictures were also included wherever available.

District profiles

This module gives various socioeconomic statistics and general information about a district of your choice.

Rural technologies

The module offers information on various technologies useful for rural development, employment generation, use of non-conventional and renewable energy.

Traditional sciences and technologies of India

This module provides information on traditional sciences and technologies of India in the field of medicine, metallurgy, architecture, agriculture and irrigation, textiles, and arts and crafts. A separate module is available to provide information about medicinal plants.

India's rich and varied heritage in Science and Technology has been universally recognized. Even up to 18th Century the technological culture in India has been identified as the forerunner of the Industrial Revolution in the West. In diverse fields of Science and Technology as Medicine, Metallurgy, Architectures, Textiles and theoretical science as Logic and Mathematics, India held a place of pride in the world before the Industrial Revolution. The survival of a large number of traditional methods and practices, especially system such as Ayurveda. Yoga and architecture, even after several thousand years, amply demonstrates the inherent strength and tenacity of traditional Indian sciences and technologies. The westernized orientation of the present system of education and the cultural invasion of the West in all channels and directions has resulted in almost of final detachment of the common man, especially the younger generation, from the Traditional Sciences and Technologies of India. This has resulted in a great loss, not only to the India scientific spirit and technological capabilities in the country, but also in affecting the developing mental planning and implementation.

However, an appropriate role the Traditional Science and Technologies can play in rural development has been recognized of late, and various efforts towards promoting the Traditional Sciences and Technologies have now begun. In this direction NIC has taken up the task of creating and disseminating a computer based database on Traditional Science and Technologies of India under the GISTNIC. This database has been classified into about 10 major subject areas and 100 minor subject areas.

GISTIC-STD code enquiry service

This allows the user to search for an STD code. By giving the first 3 characters (or the Exact name) or a place or by giving a STD code you can know the place name. You can even search by complete number (complete number as per the telephone bill and search). This makes the user to know the code of the place for which the call was made. Links to various DOT sites is also made available from which one can search for a phone number or name anywhere in India.

Companies and Industries directory

This provides product wise company profiles and addresses, while exports statistics include the trends and scenario in India Exports.

All India Primary Census Abstract 1991

The abstract of 1991 census is also made available keeping in view the importance of the census statistics. All India, statewise, districtwise, mandalwise, talukawise, townwise and villagewise details of the primary census is made available under this module.

Engineering and MBBS admissions

These are very crucial moments for the aspiring engineering and medical students community. Realizing the need and importance of this, GISTNIC project added a separate portal service. Once the EAMCET admissions start in AP student community will be anxious to know the seat vacancy position in various course and college-reservation category wise. All such vacancy position details are well updated on the EAMCET site under GISTNIC umbrella. Data will be updated right from the admission center. Every year, lakhs of people throng the GISTNIC site during admission time to know the latest status of admission. Same is the case with MBBS admissions. While engineering admission continue for a period of a least not less than 3 months with a hit rate of more than 20,000 hit per day, medical admissions with a hit rate of more than 20,000 hits per day, medical admission for a period of not more than 10 days.

Monthly Indian Economic Review

This covers the following: GNP and related indicators, Gross domestic product by industry, Net domestic product by industry, private final consumption expenditure, Gross capital formation by assets and industry, Gross capital formation by industry, Gross fixed capital formation by assets and institution, Gross fixed capital formation by industry, Gross fixed capital formation by industry, Gross Domestic saving, Net capital stock by institution, Net capital stock by industry, Net fixed capital stock by industry, Value of output, Factory incomes, Gross documents product from public sector, Gross capital formation in public sector and so on.

How to access this data?

The convergence to IT is completed. The services are being improved constantly with the new technologies and new databases. One can log on to <http://gist.ap.nic.in/> for more information on GISTNIC service.

Conclusion

In this case study, GISTNIC (General Information Service Terminal of National Information Centre), a public information service, is presented, GISTNIC predate Internet and, therefore, was a forerunner of the concept of Internet based information delivery services. In the Internet age also GISTNIC continues to be the largest source of general information on India, especially in some subject as rural technologies and traditional science and technologies of India.

E-GOVERNANCE INITIATIVE IN USA

Introduction

The President of the United States, Mr. George W. Bush had issued President's Management Agenda (PMA) for E-Government in the US in 2001. This document laid down clearly the policy, challenges, strategies, direction and future implementation plan along with past achievements in e-governance activity in the US. The details of 24 cross agency e-government initiatives and other related efforts also are given in the documents. An e-government task force was also set up to review these 24 initiatives.

Recognizing clearly the vital importance of e-governance initiatives, the Agenda in July 2001 aimed at making better use of ICT (Information and Communication Technology) investments to eliminate the massive but wasteful federal government expenditure running into billions of dollars, reduce governments paper work burden on citizens and business, and improve government responses time to citizens from weeks to minutes. Very important, a key goal was set to reduce the time taken by citizens to three click on the internet to access any government information or service. The agenda identifies internet to access any government information or service. The agenda identifies three governing principles of e-government:

- Citizen centric, instead of government centric
- Result oriented, instead of procedure oriented
- Market based, instead of regulation based.

Agendas for E-Governance

The Federal Enterprise Architecture for E-Governance

The agenda for e-governance brought out the need for Federal Enterprise

Architecture (FEA) – a comprehensive view of what government organizations does, how does it do and how IT supports it. The FEA is business focused framework that provide the office of Management and Budget (OMB), which is the agency responsible for FEA and the Federal Agencies with a methodology to monitor, analyze and control federal investments in IT. The FEA enable horizontal (Cross Federal) and

vertical (Federal, State and Local government) collaboration and communications with respect to IT investments.

Implementation of President's Management Agenda

The e-governance initiative will enable implementation of various parts of President's Management Agenda (PMA):

Budget and performance integration, strategic management of human resources, competitive procurement of services and bidding, and improved financial performance. E-payroll, e-training, (e-learning), e-clearance, and e-procurement.

Performance Measurement in E-Governance Implementations

A quarterly scoreboard is used to monitor the performance of various federal agencies engaged in e-governance implementation in their respective departmental activities. Standard requirements require the federal agencies to fulfill at least three of the following criteria:

1. Integration of citizen's one-stop service delivery through Fistgov.gov and also call centers or service centers.
2. Minimization of burden on business by re-using data previously collected or using XML to receive transmission and data
3. Inter governmental deployment of e-grants or geospatial outsourcing
4. Achieving productivity, improvements by implementing Customers Relationship Management (CRM), Supply Chain Management (SCM), Knowledge Management (KM), and Enterprise Resource Planning (ERP).

A federal Agency will be given "Green" Score if it meets all the above standards for success. "Yellow" score will be given if the agency had partially achieved this and "red" will be given if there exist significant gaps between the goals set and ground reality in the agency.

A number of agencies have shown improvement from red to yellow such as Department of Education, Department of Energy, and Department of External Affairs. NSF (National Science Foundation) upgrading itself from yellow to green.

Subsequently, 'Green' was given to a number of agencies such as Department of Housing and Urban Development, Department of Interior.

Department of Justice, Department of Agriculture, Department of Labor, Department of Transportation, Environmental Protection Agency, NASA, NSP, and Social Security Administration.

E-Government Projects, Portfolios or Categories

The e-government projects are organized into portfolio or categories as defined by the customers requirements served. The following categories are described in terms of their goals

Government to Citizens (G2C)

The objectives of (G2C) category of services is to provide one stop online access to governmental information and services to individual citizens. In this the citizens are expected to access and obtain any information in second as agents days and weeks. For examples Govbenefits.gov portal provides instantaneous access to citizens regarding information about all government program and service through a single web site within three clicks.

Government to Business (G2B)

The objective of (G2B) portfolio or category of services is to provide on stop access to business community on information pertaining to all government procedures, and rules, and regulations pertaining to them. The previous practice was making business submit multipole forms of data multiple timers to different government agencies. The objective is to provide a single window interface with the government, including all departments.

The government should re-use data submitted only once for different multiple purpose. This regulates appropriate interpretation of the same data entered in various formats. The deployment of a more effective technology will help streamline the myriad reporting requirements, as well as facilitate a more efficient means for business to interact with the government. Citizens or Business can easily find rules and regulations at www.regulation.gov even without the help of an attorney or a lawyer.

Government to Government (G2G)

The objectives of this portfolio is to enable different wings of the federal government or the State government interact effectively with each other. To achieve this, the objective of the Federal Government is to make it easier for states. So meet the reporting request while prompting the use of performance monitoring. This will definitely result in substantial saving for the State governments. The data delivery will be more accurate and time saving. An example in this direction will be Home Land Security Initiative, a security portal that will improve disaster management process, by simplifying and unifying the interaction between various government agencies as Federal, state and Local administration as well as Local Safety and Security Organizations and Personnel.

Internal Efficiency and Effectiveness (IEE)

The objectives of this portfolio is to modernize and improve the efficiency of internal process of the Federal Government in order to reduce the cost of internal administration of the Federal Government. IEE portfolio focus is to apply the best IT practice and also the bet management practice into the Federal Government operation. This aims at introducing the industry's latest best practices of Supply chain Management, Financial Managements Knowledge Management into the regular government functioning – a highly ambitions aim. However, these techniques, if introduced into the government, can revolutionize the Government functioning, speed and efficiency, eliminate all delays and improve employees satisfaction. This also includes introduction of e-learning techniques into government staff training. The e-training portal www.golearn.gov is capable of providing enhance access to high quality training and competency development for Federal Government employees.

The authentication module will deliver a solution that ensure employees authentication using multiple technologies while, at the same time protecting and respecting, the privacy and security of the users. In order words authentication provides the trust and security required for making e-government initiatives successful.

The following is a table highlighting the features of G2C and G2B:

Table G2C and G2B

G2C	G2B
Reduce the average time for citizens to find benefits and determine eligibility.	Increase the ability of citizens and business to find, view, and comment on rules and regulations.
Reduce the number of click to access relevant loan information.	Reduce burden on business by enabling online tax filing.
Increase the number of citizens filing taxes electronically.	Reduce the time to fill out export forms and locate information.
Reduce the time for citizens to find information on recreational opportunities.	Reduce time for business to file and comply with regulations.

Summarized the Features of G2G and IEE

Table: G2G and IEE

G2C	G2B
Decrease response time for jurisdictions and disciplines to respond to an emergency situations	Increases availability of training programs for government employees.
Reduce the time to verify births and death entailment information.	Reduce the average time to process clearance forms.
Increase the number of grant programs available for electronic application.	Increase use of e-travel services within each agency.
	Reduce time and overhead cost to purchase goods and services throughout the Federal government.

Underlines the Features of E-authentication.

Table: E-authentication

E-Authentication

Reduce the number of credentials by customer segment needed to interact with the Federal government.
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E-government accomplishment in USA

The following e-governance projects throw light on the e-governance initiatives in USA:

- i. **FirstGov.gov.** First Gov.gov is an American citizens' getaway to about 200 million pages of information distributed in more than 22,000 Federal and State government web sites. Named as one of the "Top 50 most incredibly useful web sites" in the entire world by Yahoo in 2002, this service is designed to provide government service within three 0clicks. The new "Three clicks" strategy has drawn dramatically large number of hits up to 30 millions.
- ii. **Volunteer.gov.** The site enables American citizens interested in volunteering to work for more than 1,00,000 opening available in government hospital and other Federal organizations.
- iii. **Recreation.gov.** The site provides citizens online access to more than the total 22,000 National parks and Recreation areas of US.
- iv. **Gov.Benefits.gov.** This site provides access to more than 400 government programs of benefits. It receives about 5,00,000 visitors per month and is listed as a 'Hot site'.
- v. **IRS free filing.** Over 80 million Americans can file their taxes online for free.
- vi. **Integrated Acquisition Environment (IAE).** This site provides information on past performance of contractors and also Federal technical data on acquisition material.
- vii. **Business.Law.gov.** The site is designed to provide small business quick access to legal and regulation information tools of compliance, the ability to perform online transactions and business related information of the government.
- viii. **Regulation.gov.** The site provides data on all government regulations. This has received 2.6 millions hits in January 2003.
- ix. **GoLearn.gov.** This is the world' most visited e-training initiative site with more than 6 millions hits, with information over 2000 e-training courses, e-books and carrier development resources. This site has more than 45,000 registered users who can receive training at negligible cost.
- x. **E-payroll.** This is a consolidated payroll from 22 different centers. More than a billion dollars re saved with this site.
- xi. **E-clearance.** This is an integrated database that will help security clearance process. 99% of all security clearance are available electronically.

Conclusion

This case study provides a clear but brief view of the IT enablement taking place in the Government of USA. The web based services offered by government of USA to the citizens are detailed in this case study.

E-GOVERNANCE IN BRAZIL

Introduction

Brazil has been taking steps towards digital government to ensure that public policies and services are more inclusive, convenient and designed to meet citizens' needs.

The use of information technology by the Brazilian Government dates back to the 1980s, but the use of the word e Government dates from 1996, with electronic services provided by the Brazilian federal Government (Ferrer, Santos, 2004). Services such as the sending of the Statement of Income Tax, information about social security and Government procurement are possible on the Internet since 1998, and in the year 2000 the Policy of e Government was defined and instituted and the Information Society Program was launched, thus consolidating and disseminating egovernment strategies, the social importance of digital inclusion as well as actions related to information technology in the country, being the eGov implemented through legal guidelines and structures in the country (Scartezini, 2004).

Scholars of the subject eGov in Brazil proved the success of the eGovernment program up to 2003, time of transition of the federal Government, when the program ceases to be a priority driven by four factors pointed out by Pinto and Fernandes (2005):

1. Change in political leadership, with diverse involvement of new employees;
2. Lack of intra bureaucratic coordination, ceasing the existence of persons responsible for the program in the different Ministries;
3. Problems in the articulation with the society, causing discontinuities in partnerships and companies that provide technological services;
4. Lack of resources for the eGov program, subsisting projects of specific sectors, however isolated from an aligned development policy.

Such factors may have interfered so that Brazil, which in 2005 ranked in the 33rd position, would end in the 45th place in the UN world classification of eGov. In the research conducted in 2010, Brazil was in the 65rd position, meaning that in the last 5 years the Brazilian eGov accumulated a loss of 32 positions in the world ranking of eGov, and in 2014 Brazil was in the 57rs position (United Nations, 2014). Despite this noticeable decline, Brazil stands out in specific initiatives, such as the Open Government Partnership, or Open Data, mentioned in the UN report as an example of good practices, by having as its objective a single point of access to public data.

The lack of online services and inadequate telecommunication infrastructure are indicated in the already mentioned report as main causes of the fall of the Brazilian classification. Currently, the Brazilian Government offers several eGov systems to the citizens.

Among the main ones the highlighted are:

1. **Federal Revenue:** services of collection of income tax, the taxpayer's tax situation, CPF and CNPJ registration, statements, among others.

2. **Federal Police:** services such as passport application, statements of criminal records, support for international adoptions, among others.
3. **Integrated System of Financial Administration of the Federal Government (SIAFI):** interests linked to the national treasure, such as provision of public expenditure.
4. **Poupa Tempo (State of São Paulo):** access to information of public services, such as document requests and starting and closing of businesses.
5. **OntoJuris Project:** provision of information on legislation in the area of intellectual property rights, consumer rights and electronic law.
6. **Public System of Digital Bookkeeping (SPED)** – presents the promotion of tax information submission, rationalization and standardization of accessory obligations for taxpayers.

Nota Fiscal Paulista Program

To understand the role of IT in the eGov initiatives in Brazil, we chose the Nota Fiscal Paulista (NFP) program, which is a pioneering initiative developed in the State of São Paulo and which is slowly being replicated in the rest of the country. Thus, we sought an in depth understanding of the NFP also showing similar programs that were implemented in other states and cities. The e-Government program in the State of São Paulo started in 1995, when the then elected State Government took over the administration with the goal of stimulating and implementing the modernization of the Public Administration.

The program involved all the Secretaries of State, brought together the governmental structure and heads of various levels in a collective effort (Filho, Agune, 2004).

The NFP program offers the following benefits (Sao Paulo State, 2010):

1. To consumers, reduced tax burden, and up to 30% of the ICMS (State Value Added Tax) collected monthly by the establishment is distributed in proportion to the purchase price to all citizens who inform their CPF or CNPJ in the act of purchase;
2. To the tax administration, it enables the expansion of the bank of taxpayers and the improvement of fiscal controls;
3. To the seller, it makes it possible the reduction of costs of paper, printing and storage of tax documents, in addition to encouraging the electronic relationship with the consumer.

We highlight as advantages of the NFP program:

1. Pioneering work: innovative project, predicting credit deposit on the bank account of the consumer, as well as conducting electronic drawings and donations;
2. Encouragement of the exercise of citizenship: participating in the program, citizens collaborate with the combat to tax evasion, exercising their citizenship;
3. Digital inclusion: the initiative stimulates the interest of the population in the use of the Internet, also offering an agreement with the "Acessa São Paulo" and the "ePoupatempo";
4. Integration with the Government: encourages the interaction of citizens, businesses and Government;

5. Ease of access: the website of the NFP program offers easily accessible interface, various manuals, consultation to claims and customer service center.

The Brazilian government can praise itself for the good use of IT technology to improve its administration and enhance interaction with citizens. In this article, you will get more information about the electronic government and its services in the country. The electronic government in Brazil had a shy beginning in 2000, with the launch of official websites of a few governmental agencies and institutions, containing only their basic information, static links and very little interaction with the users. After more than a decade of existence, the Brazilian electronic government evolved to the point of becoming an example on how the governments can better interact with citizens and [companies], improve the efficiency of internal management, cut costs and provide information and public services in a much more organized platform.

According to the last research made by the Brazilian Internet Steering Committee (CGI.br), there are currently 11.856 websites registered under the gov.br domain, what gives the size and scope achieved by the e-gov in Brazil.

If you think of the wide dimensions of the Brazilian territory and its enormous population, it gets easy to know why the electronic government contributed to the management of the public institutions, especially when it comes to the provision of public services and broadcast of information.

Overall the governmental websites obey a simple logic. If the entity belongs to the federal government, its address will end with gov.br. If it is a state institution, its address will be registered with the domain of the state it belongs. For example, the Central Bank of Brazil is a federal institution. Therefore, its address will be hcb.gov.br.

The Board of Trade of São Paulo, which is an entity under the management of that state, will have its electronic address like this: jucesp.fazenda.sp.gov.br.

The Brazilian electronic government is managed by the Administration of Information Technology Resources (SISP), organ attached the Ministry of Planning, Budget and Management.

Services Provided by the Government On Internet

Accountability and transparency

Provision of public expenditure data and documents, financial transactions, disclosure of budgets, bids, closures and cancellations and signing of contracts. This allows citizens and businesses to be aware of the government's activities as well as supervising the use of public money.

Requests

Tools that allows the citizens to make requests for public services, complain about poor and incomplete services and check schedules for public works. Because it provides the direct contact between the citizen and the public entities, this tool eliminates the need for an intermediary person, like a politician for example, who can subject the information to his interests, as a representative in the legislature.

Space for discussion

Forums in which citizens can publish their feedback about public services and campaigns, discuss or propose ideas and projects for the executive and legislative, or use of public funds. It is a way to open discussions that were restricted to the public sphere to those most interested, and also permits to achieve democratic and economic farther from public discussions of public affairs and once again, eliminating the need for intermediaries.

Ombudsman

Free spaces in which citizens can complain, request information or express their assessment of public agencies, and government services. These spaces should be free speech and should not be restricted to its users.

Registration and online services

Use of software programs and online registration for data services or perform mandatory or optional. This tool reduces cost for the public shift, manpower and time. At the same time, when only shows the alternative on-line can be exclusive.

Examples of services:

- Declaration of income
- Issuance of CPF
- Renovation of voter card
- Issuance of criminal background check
- Police report for burglary or theft
- Police report for loss of documents
- Check of lawsuits
- Check of debts related to taxes
- Check of traffic fines
- Inscription for public jobs
- Reading the state and federal official gazettes
- Issuance of passport
- Schedule for issuance of ID
- Check of grades of students from public schools
- Online library
- Public courses
- Job advertisement
- Check PIS and FGTS

A website for each state and municipality

The gov.br domain is extended to each Brazilian state creating a new domain with the acronym that they are designated. That domain will be used by the municipalities in their URL afterwards. For example, the São Paulo state domain is sp.gov.br. Bauru, a city that belongs to the São Paulo state will have its website address like this: bauru.sp.gov.br.

It is the same for the state institutions. Normally, the website of the state already has the links for the entities registered with its domain.

E-GOVERNANCE IN CHINA

Introduction

E-government has made good achievements in China and played a major role in improving administrative institutions and provision of public services. In China, the applications of e-government increased since 2001. The application of e-government in China was intended, in part, to accelerate the government's pace of implementing and using information and communications technologies (ICTs) to improve administrative efficiency and effectiveness and, through this administrative reform, to promote economic development and the administrative capacity (Hongguan Tiaokong Nengli) of the central government in China. While Chinese leaders also emphasize the importance of providing a better quality of public services to their citizens, the major goal of e-government seems to be interestingly different than in western countries. Some critical problems have arisen at the same time as in many other developing countries, such as institutional inertia, over expenditure, lack of use, lack of knowledge, lack of infrastructure, asymmetry of regional development, etc. China should consider formulating effective e-government policy as well as creating an institutional mechanism in various government departments for integrating and sharing e-government applications across provinces and local governments. This could prove especially helpful to those citizens who are living in the less developed regions and provinces lacking e-government delivery of public services.

In China, the focus of e-government has mainly been on administrative reform (Xingzheng Guanli Tizhi Gaige) which has to enter into other areas. As a consequence of a deeply centralized and often inefficient administrative management system, China has faced critical problems including over bloated structures, overstaffing, confusion between government and enterprise management, and the often unhelpful intervention of the central government in the economy (Lianjie, Chung & Thorson, 2005).

The application of e-government in China was intended, in part, to accelerate the government's pace of implementing and using information and communications technologies (ICTs) to improve administrative efficiency and effectiveness and, through this administrative reform, to promote economic development and the administrative capacity (Hongguan Tiaokong Nengli) of the central government in China (Lianjie, Chung & Thorson, 2005).

In an age of information expansion, the Chinese government has actively encouraged and invested in the growth of the Internet to capture the technology's vast commercial potential while exerting state control over the ways that this "network of networks" is accessed by Chinese citizens. The central government has encouraged the competing state-owned telecommunications providers, such as China Telecom, China Unicom, China Mobile, China Netcom, JiTong Communications Company, and China Railway Telecom,

to build their own networks. Faced with the potential political or social challenges that the new medium may bring, the Chinese Communist Party (CCP) has utilized multiple control strategies to maximize official authority.

The Hi-tech E-park

The Zhongguancun Science Park (ZSP), the biggest National Science Park in China was established in 1988, in the North-Western part of Beijing City, the capital of People's Republic of China.

The Park spans over 100 square kilometers comprising of more than 600 companies in Hi-tech areas as IT, Biology, Medicine, etc. Large multinationals such as IBM, Microsoft, Lucent, HP, and Epson have their R & D institution here. Thirty-nine prominent universities and colleges such as Beijing University and Tsinghua University are also located within the park.

The ZSP Administrative committee which is the local government body that regulates, controls and administers all activities in the park has twelve departments, each looking after functions as Company Certification, Taxation and the like.

The administration of the park formed a strategic partnership with a local private application software company, Beijing Beauty Bear Ltd which started the system study and completed the software development by the end of the year 2000.

In the new system the central database and web site allow data sharing and workflow integration among all the departments. Thus the administration functions are a single integrated body showing only one face to the public. The system includes five functions, all accessed from the web site homepage of the web site.

The following are the five functions of the system.

(i) E-application

This is a web based software that provides the applicants with all form and documents required, along with rules and regulations, and also procedure for setting up a company in the Park. After login with the user ID and password the users can fill up the files directly on the web or download the forms, fill them up on the computer, and upload them into the web. The form will depend upon the nature of the company. Almost instantaneously, the ZSP Administration committee receives the applications and begins approval procedures online. Two or three layers of approval are processed by the Administration. In each layer the Administration sends its comments/feedback/objectives to the company online. The company will make corrections online and submit immediately.

This process of submitting, correcting and resubmitting will proceed until the final approval is granted online. The applicant company will then take a printout of the approval file and submit all the original documents to ZSP verification, the licence certificate will be issued to the applicant.

As can be seen above only once there is a physical visit by the applicant to the administration office, i.e. for the final approval. Till then all the steps are automatically online.

(ii) E-registration

After obtaining approval for establishing the company, additional information is required to be provided to the administration in order to register with other department of ZSP. e.g. Statistical Bureau, Finance Bureau and Quality Control Bureau. All these now can be done through Internet.

(iii) E-reporting

Every company has to report every month to the ZSP administrations and other appropriate governments offices various information parameters such as the company revenue, tax, costs, cash flow, etc. All these can now be done online.

(iv) E-administrations

Other administrative information leading to tax cut, tax assessment, and subsidiaries, are all calculated on the web in the new systems. Financial and other information also will be submitted by the companies online.

(v) E-consulting

Government official can provide online interactive consultation regarding any of these procedures, and replies can be sent by Fax and Email.

Other features

- a. **'Red light' –reminder system.** The new system keeps track of all applications and remind the offices of how much delay and pendency is there for a given application. The 'reminder light' is clearly seen on the screen of official desk top system. First day the green light is shown. The second day yellow light and third day red light indicating the official has not completed the work on time. When the light turns red a report of delay/pendency is automatically sent to the superior officer.
- b. **Monitoring System.** Superior officers can go directly to their staff member's virtual office and check and review their work
- c. **Statistics and query monitoring System.** Statistics such as number and salary of staff, sales revenue and costs are monitored. Ad hoc, queries are also available.
- d. **OCR, VCR and HWCR.** Optical character Recognition (OCR), voice character, Reorganization (VCR), and Hand Writing Character Recognition (HWCR) are available for data entry help in Chinese language.

Benefits

A very impressive benefits list of the system can be made, Since operationalization in August 2000 several hundred thousands people have visited up to 5000. Several thousands of companies' licence and approvals of various kinds have been issued through the systems. They have all submitted their financial returns through the online web based system. Other benefits are, transparency, speed, and efficiency, added to interactive, session.

E-GOVERNANCE IN SRI LANKA

Evolution of e-Governance concept in Sri Lanka can be traced back to 1983 because the Government of Sri Lanka for the first time recognized its obligation for ICT development by creating the National Computer Policy of 1983. The Information and Communication Technology Act No. 27 of 2003 came into existence in 2003 and the Information and Communication Technology Agency of Sri Lanka was established. In 2004, "e-Sri Lanka Development Project" was initiated. It included information infrastructure building, improvement of human resources in ICT, citizen-specific service delivery, creating a modern government using ICT for social and economic development, and endorsing Sri Lanka as a destination for ICT. The e-Sri Lanka initiative expects to use ICT to develop the economy of Sri Lanka by reducing poverty and thus improving the quality of life of its citizens.

As indicated in e-Government Policy, a publication released by the Information Communication Technology Agency of Sri Lanka (ICTA) in 2010, the objectives of e-Government are the following.

- Provide more convenient and better services to citizens
- Improve participation of clients
- Use of social media to ensure better service provision
- Develop better relationships with clients
- Enhance good governance

Main features of Sri Lanka's e-Government policy E-Government policy has been designed to achieve the vision of adopting ICT to make government more efficient and effective and to create a citizen centric government. Driven by this vision the policy has addressed the following important aspects that influence the level of success in transforming traditional systems of public service delivery to e-Government.

- Enabling legal environment
- ICT management
- Information lifecycle management
- Standards applicable
- Ownership and responsibility
- Compliance requirements
- Accessibility and service delivery
- Contracts and information asset management
- ICT project continuity
- Procurement and contractual issues
- Intellectual property rights

- Communication interface
- Networking and Connectivity
- Human resource development

Some of the successful e-Government projects in Sri Lanka can be listed as follows.

- The e-Motor Revenue License project introduced in the Western Provincial Department of Motor Traffic (WPDMT) has enabled vehicle owners to renew the annual revenue license in a few minutes or obtain the license by making payments online. In formulating and implementing the e-Motor Revenue License project the WPDMT has collaborated with the Department of Motor Traffic and private sector service providers.
- The project on Digitization of Birth, Marriage and Death Certificates implemented by the Registrar General's Department has enabled residents to obtain a copy of any of the three certificates in a few minutes.
- The e-Human Resource Management project implemented by the Combined Services Division in the Ministry of Public Administration has led to a high degree of efficiency improvement, saving time and resources spent on administering annual transfers of public servants.
- The Department of Immigration and Emigration has adopted ICT successfully to improve efficiency and effectiveness of the Department. In the issue of passports to citizens and in border control activities at the international airport in Colombo the Department has been able to achieve efficiency improvements through ICT integration and training and motivation of staff

Kothamale Community Radio Internet Project

The Kothamale Radio/Internet project in Sri Lanka is an experiment in attempting to bridge the digital divide. The purpose of the Kothamale Radio/Internet project was to test and demonstrate an access model that reduce the digital divide/barriers and empower the marginalized communities in rural areas, by enabling them to enjoy benefits from ICT, to expand their knowledge base and thereby support their efforts to tackle the developmental problems themselves.

The project uses community radio as an interface between the rural community and internet through a pioneering "Radio Browsing" model. thereby introducing an indirect mass access to cyber space through a daily one-hour radio program on an interaction mode.

Example of radio/Internet Browsing

Following are some examples which describe the utility of Internet access at Kothamale community radio.

1. The village baker found various recipes from the internet to improve the quality of his products.
2. The village blacksmith browsed web sites to know more about modern implements for his trade.

3. An elderly woman visited a web site of Buddhist sacred places of India, and was very pleased as it was beyond her financial resources to visit the sacred place in person.
4. Farmer accessed the internet to find information of improved techniques of tomato growing and storage.
5. A group of local products used the Internet and found export possibilities for "Jaggery," a local product of palm.
6. An English teacher of the local school often visited with his class and downloaded lessons and games for learning English.
7. Many community members opened e-mail accounts at the radio station and used the facility to exchange e-mail with their relatives working abroad.
8. A listener visited the site to find information on bamboo craft and organized an exhibition of improved bamboo products, which he produced with the new knowledge acquired from the Internet sources.
9. A group of young people used Internet sources and networked with an organization abroad to form an environment NGO. They then mobilized resources to launch a reforestation program in their community.
10. Radio provides location-specific weather report (for the coming four days) by reading real-time satellite picture and weather forecast data from Internet sites.

Benefits

Following are the noted benefits of this project:

- An increased level of awareness concerning the benefits of ICTs.
- New livelihood ideas and model projects
- A high level of community participation and increased involvement of youth.
- Enthusiasm of local political leadership and a resurgence of commitment to the communities they serve; more animated discussion about knowledge and expansion of the communities' knowledge base.